Reading and Steering Concepts with One Direction: Representation Engineering on GPT-2 and Qwen3

github.com/javierdejesusda/tomography

TL;DR. A logistic direction reads utility and morality above a random direction on every checkpoint and truth on every Qwen3 checkpoint; the finder of the paper is weaker, a truth reader does not beat the words alone, a known-against-unknown direction does not beat the model's confidence, and utility steering that beats every random direction meets the fluency rule on only Qwen3-1.7B-Base.

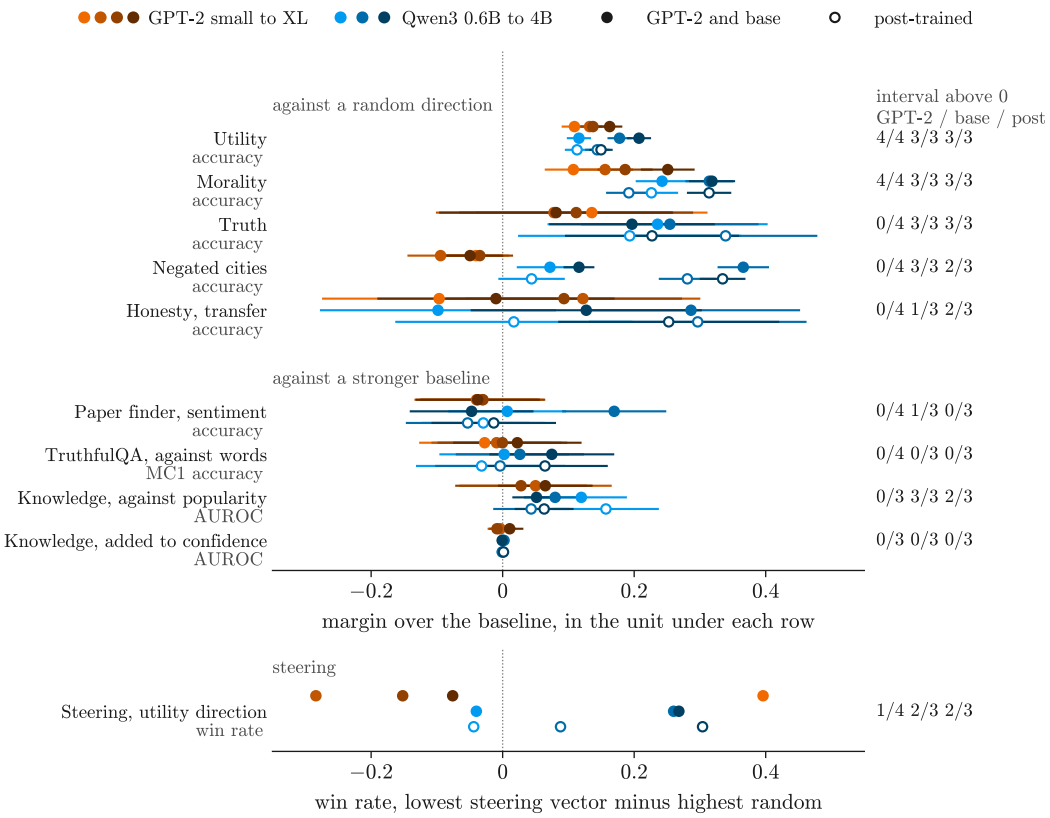


Figure 1: A direction reads utility and morality above a random direction on 4 of 4 GPT-2, 3 of 3 Qwen3 base and 3 of 3 post-trained checkpoints and truth on 0 of 4 GPT-2, 3 of 3 Qwen3 base and 3 of 3 post-trained checkpoints. Two are mostly null: the TruthfulQA reader is never above a words-only classifier, and the paper's finder beats a random direction for sentiment on 0 of 4 GPT-2, 1 of 3 Qwen3 base and 0 of 3 post-trained checkpoints. Each marker is one checkpoint's margin over its row baseline; the right column counts 95 percent intervals above zero. Steering: lowest win rate of 3 vectors minus highest of 3 random directions; no interval is stored.

Abstract

We test representation engineering (Zou et al., 2023) on four GPT-2 and three Qwen3 sizes, each as base and post-trained checkpoint, against random directions and, where the wording leaks the label, text classifiers. By the 95 percent interval, a logistic direction reads utility and morality above its random direction on every checkpoint and truth on every Qwen3 checkpoint but no GPT-2 size. The paper's finder is above a random direction for utility on every Qwen3 checkpoint, where the pairs are matched, but on fewer for sentiment and truth, where they are unrelated; matching is confounded with the concept and the data. A TruthfulQA reader is not above a classifier on the words alone, and a known-against-unknown direction, which ranks questions above popularity and text classifiers on most Qwen3 checkpoints, does not add to the confidence of the model. Steering at relative coefficient 0.1 moves judged sentiment on most checkpoints, while utility steering at relative coefficient 0.3 beats every random direction on only some. Post-training lowers held-out sentiment reading, with an interval below zero, on 2 of 3 Qwen3 sizes.

Contents: Key findings, 1 Introduction, 2 Related work, 3 Method, 4 Results, 5 Discussion and limitations, 6 Reproducibility, References, Appendix

Key findings

1. **The finder of the paper (Section 4.1).** The leading principal component of paired differences is above a random direction, with an interval excluding zero, for sentiment on 0 of 4 GPT-2, 1 of 3 Qwen3 base and 0 of 3 post-trained checkpoints, but for utility on every Qwen3 checkpoint; matching is confounded with the concept and the data. Logistic regression is above a random direction for sentiment on 4 of 4 GPT-2, 3 of 3 Qwen3 base and 3 of 3 post-trained checkpoints (Figure 3).
2. **Where a concept is read (Section 4.2).** A logistic direction reads utility and morality above its random direction on 4 of 4 GPT-2, 3 of 3 Qwen3 base and 3 of 3 post-trained checkpoints and truth on 0 of 4 GPT-2, 3 of 3 Qwen3 base and 3 of 3 post-trained checkpoints; a change of wording moves a sentiment reader by more than the noise of the test pairs on 2 of 4 GPT-2, 3 of 3 Qwen3 base and 3 of 3 post-trained checkpoints (Figure 4, Figure 1).
3. **Known and unknown questions (Section 4.3).** A known-against-unknown direction ranks questions above popularity and the text classifiers, with an interval excluding zero, on 0 of 3 GPT-2, 3 of 3 Qwen3 base and 2 of 3 post-trained checkpoints, but adds to the confidence of the model, with an interval excluding zero, on 0 of 3 GPT-2, 0 of 3 Qwen3 base and 0 of 3 post-trained checkpoints; it moves unknown questions more than random directions on 3 of 3 GPT-2 runs, 3 of 3 Qwen3 base runs and 3 of 6 post-trained runs, and a range reader is above its random control on 0 of 4 GPT-2, 2 of 3 Qwen3 base and 2 of 3 post-trained checkpoints (Figure 5, Figure 6, Figure 7).

4. **Truth (Section 4.4).** A TruthfulQA reader is above a classifier on the words alone, with an interval excluding zero, on 0 of 4 GPT-2, 0 of 3 Qwen3 base and 0 of 3 post-trained checkpoints (Figure 8).
5. **Steering (Section 4.5).** Steering at relative coefficient 0.1 moves the judged sentiment on 4 of 4 GPT-2, 3 of 3 Qwen3 base and 2 of 3 post-trained checkpoints and keeps the fluency rule (neutral-text perplexity ratio at most 1.5 and distinct-1 at least 0.15) on 3 of 4 GPT-2, 3 of 3 Qwen3 base and 3 of 3 post-trained checkpoints; utility steering at relative coefficient 0.3 beats every random direction on 1 of 4 GPT-2, 2 of 3 Qwen3 base and 2 of 3 post-trained checkpoints (Figure 9).
6. **Post-training (Section 4.6).** Held-out sentiment reading is lower after post-training, with an interval below zero, on 2 of 3 Qwen3 sizes, and 3 of 8 other seed-paired differences exclude zero (Figure 10).

1 Introduction

Zou et al. [1] propose representation engineering: find a direction for a concept from the activations of contrasting inputs, read the concept by projecting new activations on it, and control the model by adding it to the residual stream. The recipe rests on the linear representation hypothesis, the view that concepts correspond to directions in activation space [2]. The paper reports reading and control results on LLaMA-2 chat models for truthfulness, honesty, utility, morality, emotions and other concepts.

We run the recipe on smaller open models where every step can be checked: GPT-2 [3] at four sizes and Qwen3 [4] at three sizes, each as a base and a post-trained checkpoint. Our readers are logistic regression unless a figure names the finder of the paper, because that finder is above a random direction by its interval for only some concepts and roles (Section 4.1). For each question we name the baseline that could produce the result without the direction: a random direction fitted with the same procedure, a classifier on the text (for the readings, TruthfulQA, knowledge and approximate answers, where the wording can give the label away), the popularity of the subject, or the confidence of the model. Three terms recur. The margin is the accuracy of a reader minus that of a random direction of the same run; the relative depth is a layer index divided by the number of layers; and the relative coefficient is the steering strength in units of the mean residual norm at the layer. Figure 1 summarises the answers.

Our contributions are:

1. **Baseline-relative readings (Section 3).** Every reading is reported against a random direction fitted with the same procedure and, where the wording can give the label away, against a text baseline, with counts per role and never pooled.
2. **A test of the finder (Section 4.1).** The finder of the paper is compared with logistic regression on matched pairs (utility) and on unrelated pairs (sentiment and truth).
3. **Two families, base and post-trained (Section 4.6).** Four GPT-2 sizes and three Qwen3 sizes, each Qwen3 size as a base and a post-trained checkpoint, so a difference between roles is measured, not assumed.

4. **Known and unknown questions (Section 4.3).** A known-against-unknown direction is compared with popularity, text classifiers and the confidence of the model, its steering with random directions of the same norm, and a range direction for approximate answers with a random control.
5. **A page rebuilt from the result files (Section 6).** Every number comes from the committed results and is recomputed when the page is built.

2 Related work

Fitting a classifier on hidden states to read a property is probing [5] [6]. A probe can succeed because it learns the task rather than reads it, which control tasks are built to expose [7]; our control is a random direction fitted with the same orientation rule and, where the wording can give the label away, a classifier on the text alone. Representation engineering [1] reads a concept with the first principal component of differences between paired activations and compares that finder with others on utility in its Section 5.1; several finders recover a similar sentiment direction in GPT-2 and Pythia [8], and on a shared benchmark difference-in-means is the best concept detector, prompting beats every method for steering (followed by finetuning) and sparse autoencoders are not competitive on either [9].

Truth has been read from hidden states with supervised probes [10], with mass-mean probes whose directions transfer across datasets of true and false statements [11], and without labels by contrast-consistent search [12]; such probes can fail on negated statements [13], which our negated sets test.

Adding a direction to the residual stream follows activation addition [14] and contrastive activation addition [15]; inference-time intervention shifts the outputs of selected attention heads along probe directions [16], and one direction mediates refusal in chat models [17]. Steering vectors are unreliable for many inputs and concepts [18].

Language models can report whether they know an answer [19], answerability can be decoded from their hidden states [20], probes of the subject representation predict which PopQA facts a model knows better than popularity does [21], and directions for known and unknown entities steer refusal in chat models [22]. Popularity itself predicts factual recall on PopQA [23].

Model behaviour is sensitive to formatting choices in the prompt [24], which motivates our wording test. Du et al. [25] compare base and post-trained models on knowledge, truthfulness, refusal and confidence; our comparison is limited to the released Qwen3 pairs.

3 Method

Table 1 lists the checkpoints, Figure 2 shows the pipeline and Table A2 defines the symbols and terms; Appendix B gives every design detail.

Table 1: The 10 checkpoints range from 124M to 4.02B parameters, in four GPT-2 sizes and three Qwen3 sizes with a base and a post-trained model each. Parameters include the embeddings, the positional table of GPT-2 and every norm, and are computed from the stored model configuration. Layers are decoder blocks. The Hub repository, the pinned revision and the license of each checkpoint are in Table A1. Post-trained Qwen3 checkpoints are used with their chat template and thinking disabled; they differ from their base checkpoint in post-training data, prompt format and distillation from a larger model.

Model	Role	Parameters	Layers	Hidden size
GPT-2 small	GPT-2	124,439,808	12	768
GPT-2 medium	GPT-2	354,823,168	24	1024
GPT-2 large	GPT-2	774,030,080	36	1280
GPT-2 XL	GPT-2	1,557,611,200	48	1600
Qwen3-0.6B-Base	base	596,049,920	28	1024
Qwen3-0.6B	post-trained	596,049,920	28	1024
Qwen3-1.7B-Base	base	1,720,574,976	28	2048
Qwen3-1.7B	post-trained	1,720,574,976	28	2048
Qwen3-4B-Base	base	4,022,468,096	36	2560
Qwen3-4B	post-trained	4,022,468,096	36	2560

3.1 Reading a concept

A reading experiment builds pairs of inputs that differ in the concept: two phrases of opposite sentiment, two versions of a scenario with different utility or morality, a true and a false statement. Each input is wrapped in the template of the paper for the concept, and we read h(x, ℓ), the residual stream at the last token of the input x after layer ℓ. A finder turns the training pairs into one unit direction v(ℓ) per layer; unless a figure names another finder, it is logistic regression on the paired activations. The direction is oriented with the training labels, so that the positive member of a training pair tends to project higher. A held-out pair is read correctly when its positive member projects higher than its negative member, and ties count one half; the accuracy is the share of held-out pairs read correctly, so a direction with no information scores 0.5 and no threshold is involved.

The layer chosen on validation is picked on pairs that the reported score does not use. The main sentiment run is the exception: it chooses its layer on the same 256 SST-2 validation pairs it is scored on, so its value is optimistic. We therefore report sentiment from the sample-size run at 64 training pairs, which scores a separate report split of 214 pairs, and keep the optimistic run in Table A4 only.

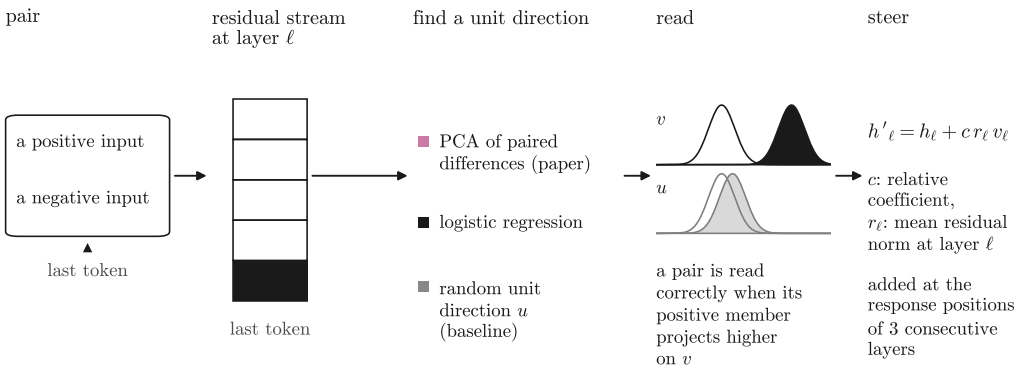


Figure 2: The recipe: find a direction, read with it, and steer with it. A pair of inputs that differ in the concept is passed through the model; a finder turns the residual-stream activations of the last token at layer ℓ into a unit direction v; a pair is read correctly when its positive member projects higher on v, so a constant score sits at 0.5; steering adds c r v to the residual stream. A random unit direction, oriented on the training pairs like the reader, is the baseline of the readings. The densities in the read stage are illustrative.

3.2 Checkpoints and data

GPT-2 [3] small, medium, large and XL, and Qwen3 [4] 0.6B, 1.7B and 4B, each as the base checkpoint and the post-trained one. A post-trained checkpoint is used with its chat template and thinking disabled. Sentiment is read on phrases of the Stanford Sentiment Treebank [26], utility and morality on ETHICS [27], and truth on the true and false statements of the representation engineering repository [1], tested out of distribution on the Geometry of Truth sets [11]. TruthfulQA [28], PopQA [23] and Wikidata quantities [29] give the question sets of the baselines, the knowledge directions and the approximate answers; generated text is scored by the positive-class probability of SiEBERT [30]. Datasets, judges and revisions are in Table A1.

3.3 Baselines

Every reading run also scores random unit directions u(ℓ) at every layer, oriented on the training labels exactly like the reader. Because the orientation uses the labels, a random direction can score far above 0.5 when the concept dominates the activations (up to 0.859, on Qwen3-4B-Base, in the finder comparison), so every reading is reported as its margin over the random direction of the same run, not over 0.5; the two share a layer except in the finder comparison, which reads each at the layer chosen on its own select split. Where the wording of an input can give the label away (the sentiment, utility, morality and truth readings, TruthfulQA, knowledge questions, approximate answers), we add classifiers on the words and on the characters of the input, fitted on the same training items without the model (Table 2 gives them as words alone); for knowledge we add the popularity of the subject, its monthly Wikipedia page views, and the confidence of the model, which stacks five output signals. The steering comparisons use three random directions of the same norm.

3.4 Steering

Steering adds c r(ℓ) v(ℓ) to the residual stream at three consecutive layers (for the sentiment run only; the utility steering run steers at 40, 55 and 70 percent of the depth), at every position of the response, where v(ℓ) is the unit direction, r(ℓ) is the mean residual norm at layer ℓ on calibration text and c is the relative coefficient. The judged score of a checkpoint is the mean SiEBERT positive-class probability of its completions. We call steered text fluent only when two conditions hold, the fluency rule: the neutral-text perplexity ratio at most 1.5 and distinct-1 at least 0.15 (the share of distinct tokens in the completions [31]). Three random directions of the same norm are the reference for the win rate of the utility test, not 0.5.

3.5 Intervals and counts

Every reading interval on this page is computed from the test size: a 95 percent Wilson interval [32] for an accuracy, a conservative interval for a difference between two accuracies, and the approximation of Hanley and McNeil [33] for an AUROC. No interval includes the variation between independent runs, because each cell is one run. Counts of checkpoints are given per role (GPT-2, Qwen3 base, Qwen3 post-trained) and never pooled; they are descriptive, and no correction for the number of comparisons is applied.

4 Results

4.1 Does the finder of the paper find the concept?

The finder of the paper is above a random direction by the 95 percent interval for utility on every Qwen3 checkpoint, where the pairs are matched, and on fewer checkpoints for sentiment and truth, where they are unrelated, while logistic regression is above one for sentiment on every checkpoint.

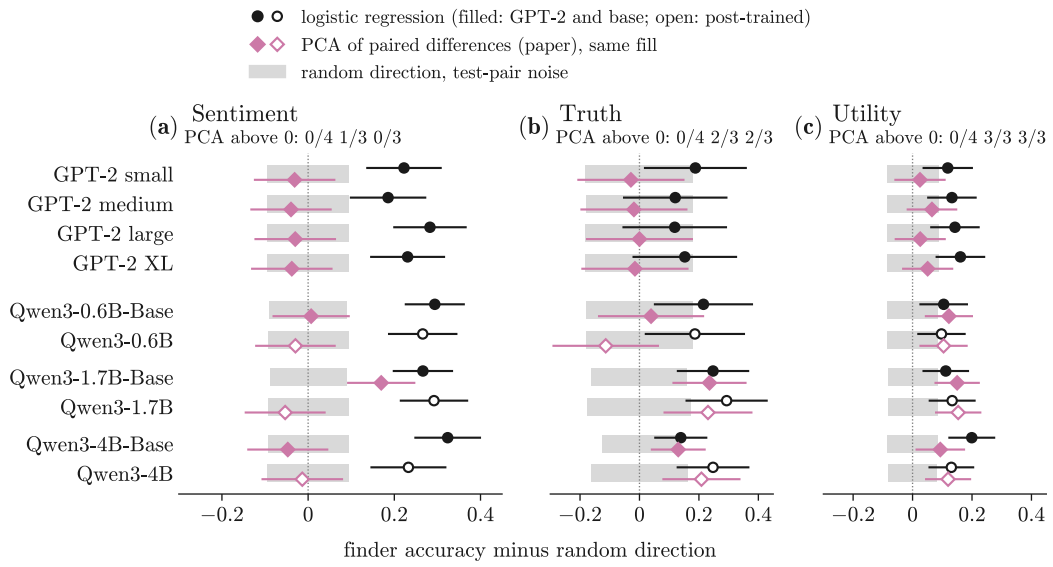


Figure 3: On unrelated sentences with opposite labels, the leading principal component of paired differences is above a random direction, with an interval excluding zero, for sentiment on 0 of 4 GPT-2, 1 of 3 Qwen3 base and 0 of 3 post-trained checkpoints and for truth on 0 of 4 GPT-2, 2 of 3 Qwen3 base and 2 of 3 post-trained checkpoints. On utility (matched pairs) it is above it on 3 of 3 Qwen3 base and 3 of 3 post-trained checkpoints by interval and on GPT-2 by point estimate only (4 of 4, by 0.025 to 0.065), but matching is confounded with the concept. Logistic regression is above it for sentiment on 4 of 4 GPT-2, 3 of 3 Qwen3 base and 3 of 3 post-trained checkpoints. Each marker is a finder's report-split accuracy minus that of a random direction, at the layer its select split chose (filled: GPT-2 and Qwen3 base; open: post-trained); whiskers are conservative 95 percent intervals over the test pairs (214 sentiment, 59 truth and 256 utility pairs); the grey band is test-pair noise. The line above each panel counts checkpoints with the paper finder's interval above zero. Layer curves: Figure A3, Figure A4; all six finders: Table A3.

The paper compares finders on utility, where every pair is two versions of one scenario. On sentiment and truth our pairs are unrelated inputs with opposite labels, the setting of most probing work. Matching is confounded with the concept and the data, so the comparison does not show that matching is what matters. The layer curves are in Figure A3 and Figure A4 and every finder in Table A3 (Appendix C).

4.2 Where and how well is a concept read?

A logistic direction reads utility and morality above its random direction on every checkpoint and truth on every Qwen3 checkpoint but no GPT-2 checkpoint, and a change of the wording of the template moves a sentiment reader by more than the noise of the test pairs on every Qwen3 checkpoint and on two of the four GPT-2 checkpoints.

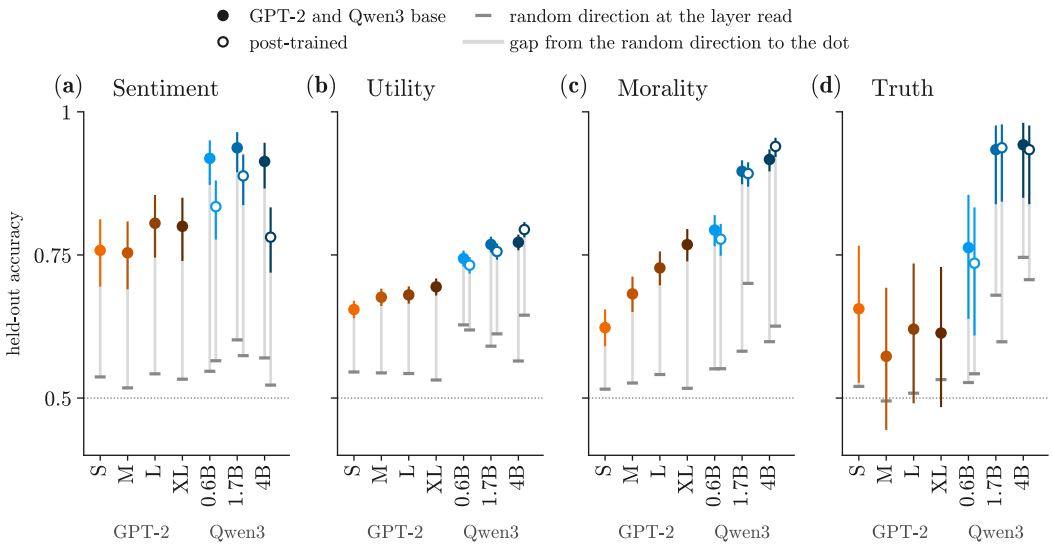


Figure 4: The held-out accuracy of a logistic direction is above that of its random direction, with an interval excluding zero, for sentiment, utility and morality on 4 of 4 GPT-2, 3 of 3 Qwen3 base and 3 of 3 post-trained checkpoints and truth on 0 of 4 GPT-2, 3 of 3 Qwen3 base and 3 of 3 post-trained checkpoints. Held-out accuracy at the layer read against checkpoint size (S to XL: GPT-2; 0.6B to 4B: Qwen3; filled for GPT-2 and base, open and shifted right for post-trained), with 95 percent Wilson intervals for the test pairs (214 sentiment, 4,808 utility, 983 morality and 59 truth pairs) and the random direction at the same layer as a grey tick; the sentiment dots come from the sample-size run at 64 training pairs, read at its own layer. GPT-2 slopes (not drawn) are least-squares over four sizes against log parameters, with a t interval: across the 4 GPT-2 sizes accuracy rises by 0.034 for utility (0.013 to 0.055) and 0.132 for morality (0.128 to 0.136) per tenfold in parameters (95 percent intervals); the sentiment slope (0.048, −0.050 to 0.146) and the truth slope (−0.027, −0.231 to 0.177) have intervals that include zero. Onsets: Figure A1.

Table 2: Reader accuracy at the layer chosen on validation, next to the random direction of the same run at that layer. An asterisk marks a reader whose margin over the random direction has a conservative 95 percent interval above zero for the test pairs: 214 sentiment pairs (sample-size run, 64 training pairs), 4,808 utility, 983 morality and 59 truth pairs, and 59 plain true and false statements for honesty, read by a direction fitted on the paper's honesty persona prompts with no refit. Margins with their intervals, the layers and the in-template honesty values are in Table A4. The last two rows give the words alone, a text classifier on word or character n-grams of the pair texts, with no model: each value is the mean over the training sets (20 for the held-out sentiment split and 10 for the others) and its interval covers the sampling of the test pairs only, not the spread over training sets. The honesty baseline is a classifier fitted on the true and false labels of the facts, not on the persona labels of the reader, so it is a baseline for the transfer pairs only.

	Sentiment (held out)		Utility		Morality		Truth		Honesty (transfer)	
Model	Reader	Random	Reader	Random	Reader	Random	Reader	Random	Reader	Random
GPT-2 small	0.758*	0.537	0.655*	0.546	0.623*	0.516	0.656	0.520	0.386	0.483
GPT-2 medium	0.754*	0.518	0.676*	0.544	0.682*	0.526	0.573	0.495	0.598	0.476
GPT-2 large	0.805*	0.542	0.680*	0.543	0.727*	0.541	0.620	0.508	0.566	0.473
GPT-2 XL	0.800*	0.533	0.694*	0.532	0.768*	0.517	0.614	0.532	0.492	0.502

	Sentiment (held out)		Utility		Morality		Truth		Honesty (transfer)	
Model	Reader	Random	Reader	Random	Reader	Random	Reader	Random	Reader	Random
Qwen3-0.6B-Base	0.919*	0.547	0.744*	0.628	0.793*	0.551	0.763*	0.527	0.456	0.554
Qwen3-0.6B	0.834*	0.565	0.732*	0.619	0.778*	0.551	0.736*	0.542	0.529	0.512
Qwen3-1.7B-Base	0.937*	0.602	0.768*	0.591	0.896*	0.582	0.934*	0.680	0.778*	0.492
Qwen3-1.7B	0.888*	0.574	0.756*	0.612	0.892*	0.700	0.937*	0.598	0.781*	0.485
Qwen3-4B-Base	0.913*	0.570	0.772*	0.565	0.917*	0.598	0.942*	0.746	0.658	0.531
Qwen3-4B	0.781*	0.523	0.794*	0.645	0.940*	0.626	0.934*	0.707	0.758*	0.505
Words alone (word n-grams)	0.593 [0.526, 0.656]		0.520 [0.506, 0.534]		0.517 [0.486, 0.548]		0.568 [0.441, 0.686]		0.603 [0.476, 0.718]	
Words alone (character n-grams)	0.636 [0.570, 0.698]		0.525 [0.510, 0.539]		0.527 [0.495, 0.558]		0.524 [0.399, 0.646]		0.585 [0.458, 0.702]	

The margin over a random direction exceeds 0.1 for sentiment from a relative depth of 0.021 to 0.167 on GPT-2, 0.036 to 0.083 on Qwen3 base and 0.036 to 0.056 on post-trained checkpoints, and later for utility, morality and truth; across the GPT-2 sizes accuracy grows with size for utility and morality, and the truth reader on GPT-2 is not separable from its random direction at 59 test pairs. The words alone read sentiment (0.593 with word n-grams, 0.636 with character n-grams on the held-out split) and utility (0.520, 0.525) with an interval above chance, and do not separate from chance for morality, truth and honesty (Table 2). The wording study refits the sentiment reader on changed templates (Figure A5, Figure A6, Table A7; Appendix D). Layer curves and every reading with its layer are in Appendix C and Figure A1.

4.3 Known and unknown questions

A known-against-unknown direction ranks questions above popularity and text classifiers on most Qwen3 checkpoints but adds at most a small amount to the confidence of the model, and a range reader is above its random control on only some checkpoints.

Reading. Figure 5 compares the direction with popularity, text and the confidence of the model on PopQA within relation: it ranks questions above the strongest of popularity and the two text classifiers, with an interval excluding zero, on 0 of 3 GPT-2, 3 of 3 Qwen3 base and 2 of 3 post-trained checkpoints, and its increment over the confidence of the model has an interval covering zero on 3 of 3 GPT-2, 3 of 3 Qwen3 base and 3 of 3 post-trained checkpoints, at most 0.032 (GPT-2), 0.006 (Qwen3 base) and 0.004 (post-trained) at its upper end, while a probe fitted on the correct and wrong labels has an interval above zero on 4 of 4 GPT-2, 3 of 3 Qwen3 base and 3 of 3 post-trained checkpoints; GPT-2 small has no knowledge-direction run, so the denominators differ. On Qwen3-4B-Base a classifier on the words or characters comes within a small margin of the direction or beats it on 9 of 14 question types (Figure A8).

Steering. Adding the direction moves the outcome on unknown questions (the next-token entropy for GPT-2 and base checkpoints, the abstention rate for post-trained ones) more than each of three random directions of the same norm on 3 of 3 GPT-2 runs, 3 of 3 Qwen3 base runs and 3 of 6 post-trained runs, and more than on known questions on 3 of 3 GPT-2 runs, 3 of 3 Qwen3 base runs and 4 of 6 post-trained runs; known questions move at least as much as unknown ones on 2 of 3 GPT-2 runs, 2 of 3 Qwen3 base runs and 0 of 6 post-trained runs (Figure 6; the measures are in Appendix G).

Approximate answers. Asked for a quantity whose sourced answer is a range, Qwen3 base checkpoints write a range for 83 to 99 percent of the items and post-trained ones for 28 to 68 percent when the last example of the prompt is a range, and, told to give one exact number, for at most 21 percent (base) and 1 percent (post-trained). A range reader fitted on heights and wingspans is above its random control on held-out durations on 0 of 4 GPT-2, 2 of 3 Qwen3 base and 2 of 3 post-trained checkpoints, and above a classifier on the question words, with the interval of the difference above zero, on 0 of 4 GPT-2, 0 of 3 Qwen3 base and 0 of 3 post-trained checkpoints (Figure 7).

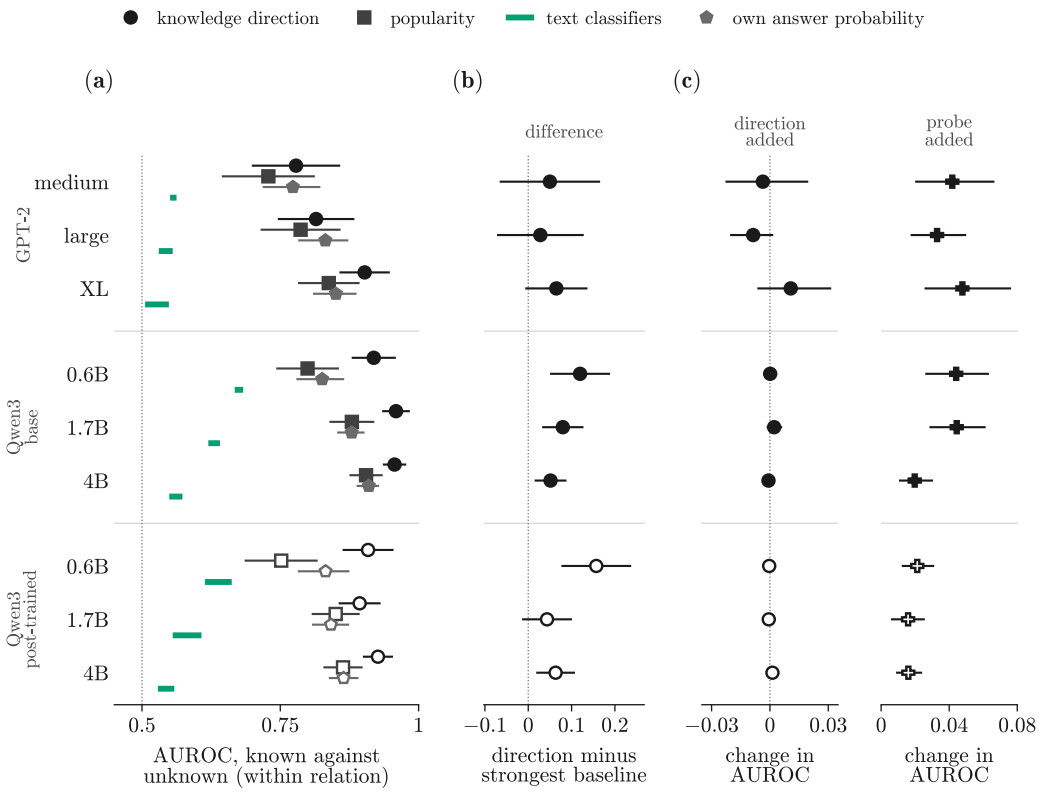


Figure 5: A known-against-unknown direction ranks questions above the strongest of popularity and the two text classifiers, with an interval excluding zero, on 0 of 3 GPT-2, 3 of 3 Qwen3 base and 2 of 3 post-trained checkpoints, and adds at most 0.032 (GPT-2), 0.006 (Qwen3 base) and 0.004 (post-trained) to the confidence of the model at the upper end of its interval. (a) Within-relation AUROC on PopQA (open markers: post-trained): direction (circle), popularity (square), range of the two text classifiers (green bar) and its first answer token probability (pentagon). GPT-2 0.778 to 0.902 against 0.728 to 0.838, 0.511 to 0.557 and 0.773 to 0.850; Qwen3 base 0.919 to 0.959 against 0.799 to 0.905, 0.555 to 0.678 and 0.826 to 0.910; Qwen3 post-trained 0.893 to 0.927 against 0.752 to 0.863, 0.534 to 0.657 and 0.832 to 0.864. (b) Difference in AUROC, direction minus the strongest of popularity and the two text classifiers, conservative interval. (c) Change in AUROC for wrong answers when the direction, or a probe fitted on the correct and wrong labels, is added to model confidence; the interval covers zero on 3 of 3 GPT-2, 3 of 3 Qwen3 base and 3 of 3 post-trained checkpoints. See Appendix G.

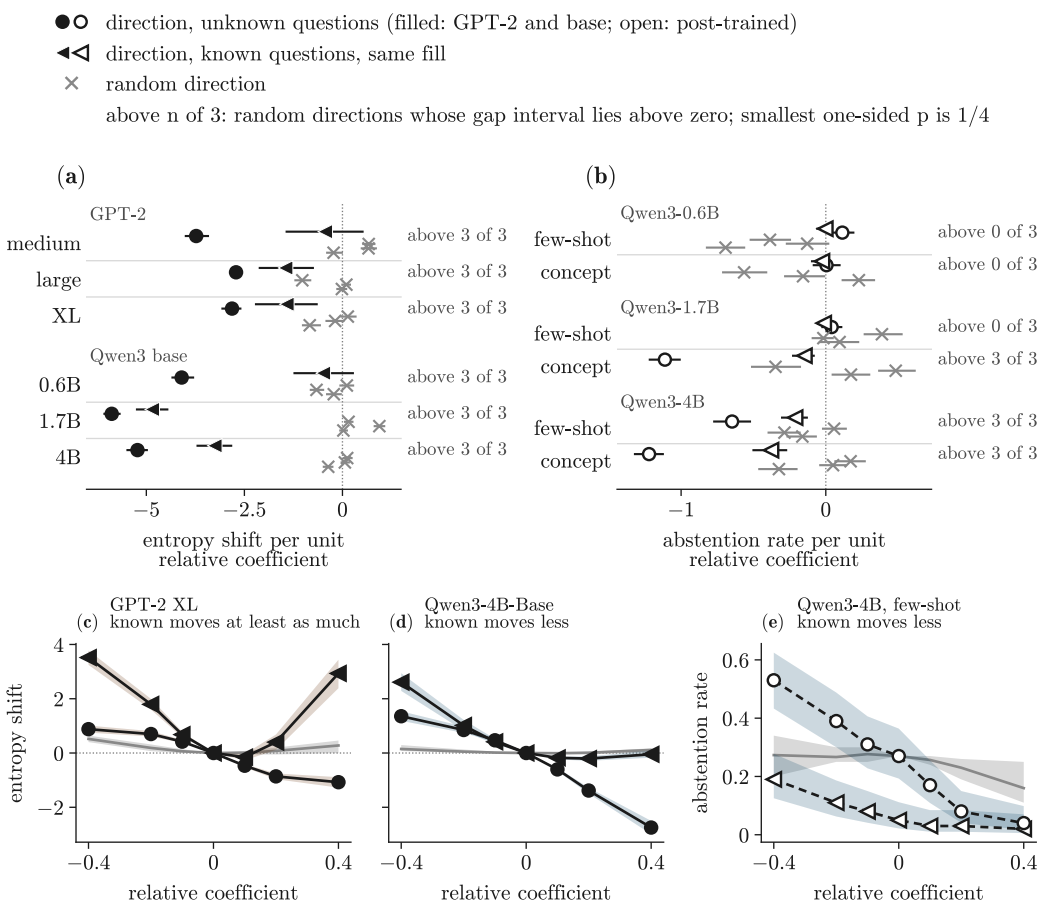


Figure 6: Adding the knowledge direction moves unknown questions more than each of 3 random directions of the same norm on 3 of 3 GPT-2 runs, 3 of 3 Qwen3 base runs and 3 of 6 post-trained runs, yet by the largest shift of the mean dose curve (c to e), which the slopes in (a, b) do not show, known questions move at least as much as unknown ones on 2 of 3 GPT-2 runs, 2 of 3 Qwen3 base runs and 0 of 6 post-trained runs. (a) Slope of the entropy shift per unit relative coefficient on unknown questions (circle) and known questions (left triangle) of GPT-2 and base checkpoints, against 3 random directions (grey crosses), with 95 percent intervals that resample the questions. (b) The same for the abstention rate of post-trained checkpoints (few-shot and concept prompt); outcomes are never pooled. "above n of 3" counts the random directions beaten. (c to e) Dose curves of GPT-2 XL, Qwen3-4B-Base and Qwen3-4B (few-shot) against the mean of the 3 random directions on unknown questions with its interval (grey band) and the direction's interval (tint). Rows run by size; GPT-2 small has no run. All runs: Figure A9; by question type: Figure A10.

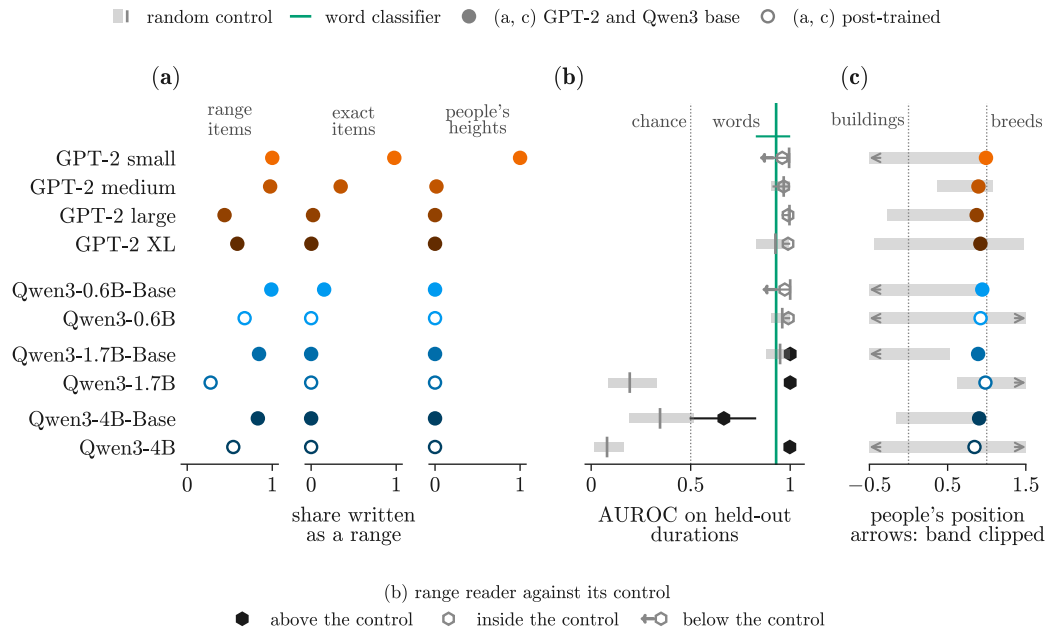


Figure 7: Qwen3 writes a range for 83 to 99 percent of range items (base) and 28 to 68 percent (post-trained), and a range reader is above the interval of its random control on 0 of 4 GPT-2, 2 of 3 Qwen3 base and 2 of 3 post-trained checkpoints. (a) Share of few-shot answers written as a range on range items, exact items and 120 people's heights. Wilson 95 percent intervals lie inside the markers, at most 10.2 (GPT-2), 7.8 (Qwen3 base) and 10.3 (post-trained) points wide. Exact items, heights: at most 15 percent of exact items, none of the heights (Qwen3 base), none of either (post-trained); GPT-2 small 100 and 98 percent on range and exact items. (b) AUROC on durations of a range reader fitted on heights and wingspans, with its random control (grey bar and tick: bootstrap interval and mean of 5 random directions) and a word classifier (green line 0.929, whisker 0.827 to 1.000); a filled hexagon is above the control's interval, an open hexagon inside it, a left arrow below it. (c) People's position between buildings (0) and breeds (1) against the random band, the range of the control's position over its 7 training sets. See Appendix H.

Every dose curve is in Appendix G and the values behind the range reader in Appendix H.

4.4 Does a truth reader beat the words alone?

A TruthfulQA reader is not above a classifier on the words alone.

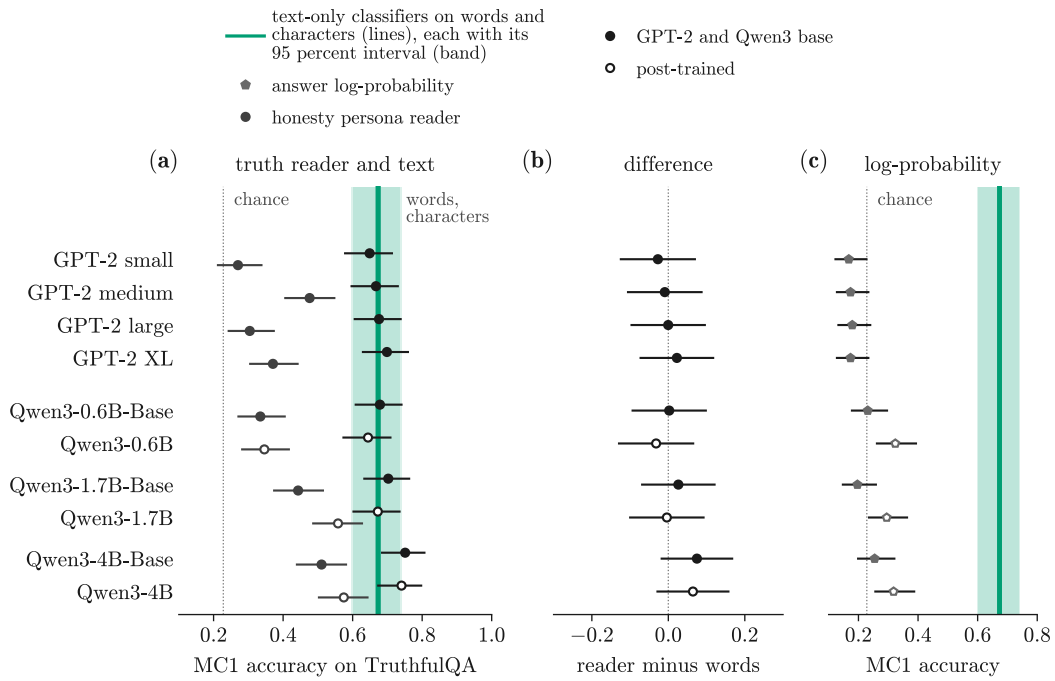


Figure 8: A truth reader does not beat words alone on TruthfulQA: its MC1 accuracy lies inside the span of the 95 percent intervals of two text-only classifiers (0.597 to 0.742; their accuracies are 0.676 and 0.671) on 4 of 4 GPT-2, 2 of 3 Qwen3 base and 3 of 3 post-trained checkpoints, while the honesty persona reader reaches 0.270 to 0.476 on GPT-2, 0.334 to 0.511 on Qwen3 base and 0.346 to 0.575 on post-trained, against a chance level of 0.228. (a) MC1 accuracy of the truth reader (circle) and of the honesty persona reader (dark grey circle; open for post-trained) at the layer chosen on validation, with 95 percent Wilson intervals over 173 questions; each text-only classifier is a green line (the two nearly coincide) with its own interval as a band; the dotted line is the chance level of 0.228. (b) The reader minus the more accurate classifier, with the conservative 1.96-standard-error interval of a difference of two proportions over the same 173 questions; intervals that exclude zero: 0 of 4 GPT-2, 0 of 3 Qwen3 base and 0 of 3 post-trained checkpoints. (c) The answer log-probability with a Wilson interval beside the band. Negation statements are in Figure A7.

TruthfulQA can be answered in part from the answer text alone, and negation flips the label without changing the topic: Figure 8 pairs the reader with two text classifiers, and at least one negated set falls below chance with its interval on 3 of 4 GPT-2, 0 of 3 Qwen3 base and 2 of 3 post-trained checkpoints (Figure A7). The readers and the honesty transfer are in Appendix E.

4.5 Does steering change the output without breaking it?

Steering at relative coefficient 0.1 moves the judged sentiment on most checkpoints and keeps the fluency rule on most, but at relative coefficient 0.3 the steered sentiment text meets the rule on none, and utility steering beats every random direction on only some checkpoints.

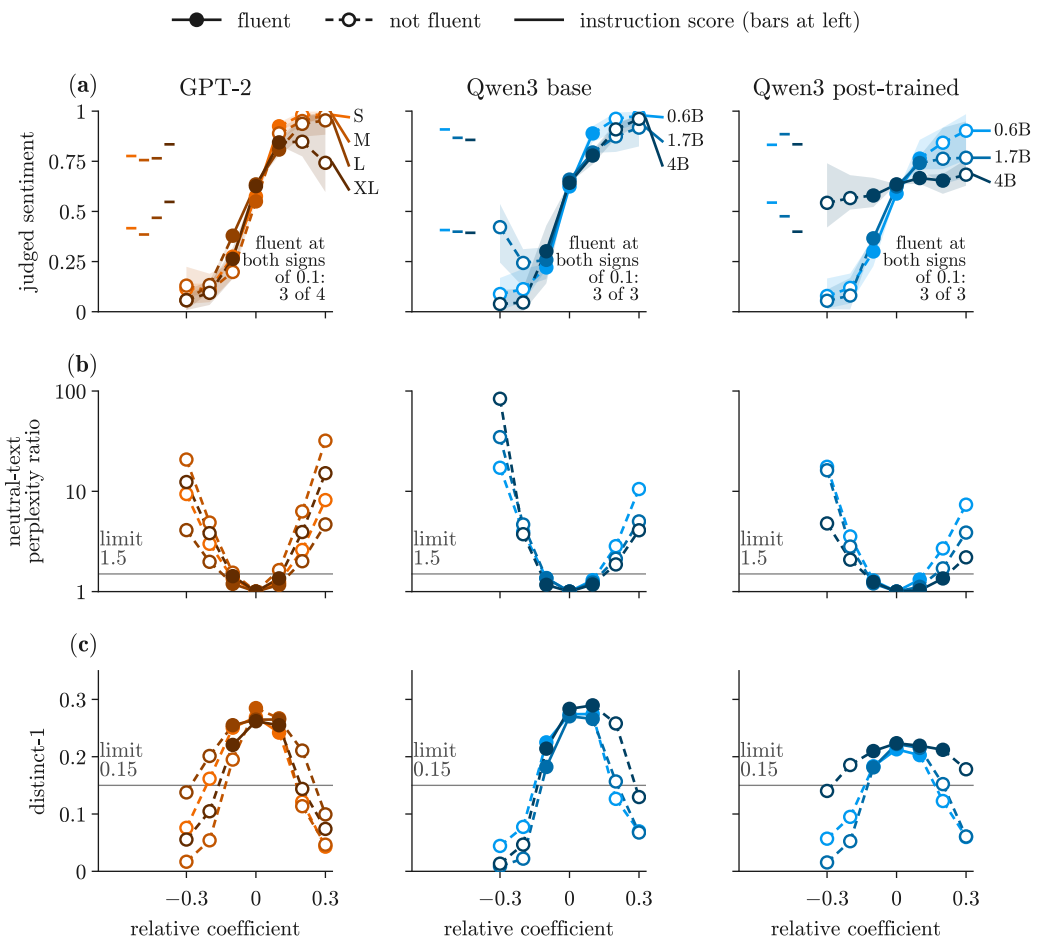


Figure 9: Steering moves the judged sentiment, less on post-trained checkpoints. A checkpoint moves when all 3 steering vectors move the score past the unsteered value at both signs of 0.1 on two sentiment judges: 4 of 4 GPT-2, 3 of 3 Qwen3 base and 2 of 3 post-trained checkpoints. The fluency rule holds at both signs of 0.1 on 3 of 4 GPT-2, 3 of 3 Qwen3 base and 3 of 3 post-trained checkpoints, and at 0.3 or −0.3 on 0 of 4 GPT-2, 0 of 3 Qwen3 base and 0 of 3 post-trained checkpoints. (a) Mean judged sentiment against the relative coefficient, one line per checkpoint, with the range over the 3 steering vectors (band) and the score under a positive or negative instruction (bars at left). Filled markers satisfy the fluency rule (perplexity ratio at most 1.5, distinct-1 at least 0.15); a segment touching an open marker is dashed. (b) Perplexity ratio, steered to unsteered model, on 200 held-out WikiText-103 passages. (c) Distinct-1 of the completions. The rule uses the perplexity of neutral text, not of the completions (Table A10 gives the scores, ratios and judge agreement). It has no random-direction arm; the utility steering run uses random directions (Figure 1).

The comparison with random directions is in Figure 1 (last row), and Table A10 adds the completion perplexity (Appendix F).

4.6 What does post-training change?

Post-training lowers held-out sentiment reading, with an interval below zero, on 2 of 3 Qwen3 sizes, and on the other readings most intervals include zero, which is not evidence of no change.

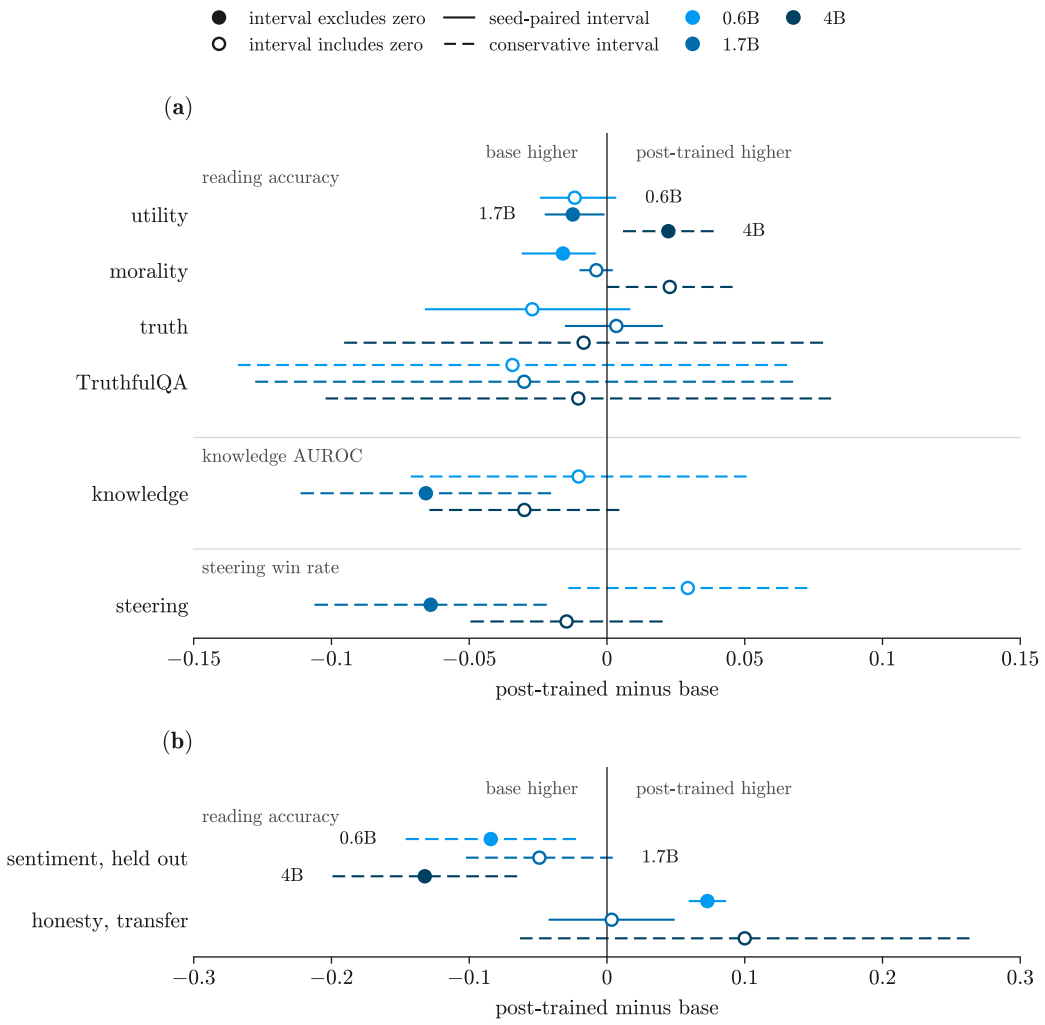


Figure 10: Post-training lowers held-out sentiment reading; on the other readings most intervals include zero, which is not evidence of no change. Sentiment falls by 0.049 to 0.132, with an interval below zero on 2 of 3 sizes; utility, morality, truth, honesty transfer and TruthfulQA move by at most 0.100, with an interval excluding zero only for utility on 1.7B (−0.012), utility on 4B (+0.022), morality on 0.6B (−0.016) and honesty transfer on 0.6B (+0.073); the knowledge AUROC and the steering win rate fall with an interval below zero on 1.7B only. Post-trained minus base checkpoint for each Qwen3 size (0.6B, 1.7B and 4B, light to dark). A marker is filled where the interval excludes zero, open otherwise. A solid interval is the seed-paired 95 percent bootstrap interval that Table A13 stores (10 seeds, 0.6B and 1.7B); a dashed interval is a conservative 95 percent interval, used for 4B and the measures without seeds. The axes differ between the panels: (a) shows the rows whose intervals stay within 0.150, (b) the rows with wider intervals, on −0.300 to 0.300. A post-trained checkpoint also differs from its base in prompt format and distillation, so the contrast is with this recipe (Section 5).

Figure 10 puts every measure of the earlier sections side by side as post-trained minus base for the three Qwen3 sizes; the stored per-seed comparisons are in Table A13 (Appendix I).

5 Discussion and limitations

A direction reads a concept better than chance far more often than it beats the right baseline: orientation on the labels lifts a random direction, the wording of the input often carries the label, and the confidence of the model already holds what a knowledge direction reads, so a reading claim needs its random direction and, where the text can carry the label, a text baseline (Figure 1). Steering moves the judged sentiment at small coefficients and breaks fluency at larger ones (a perplexity on neutral text above 1.5 times the unsteered model, or a distinct-1 below 0.15; Section 4.5), and post-training lowers held-out sentiment reading, with an interval below zero, on 2 of 3 Qwen3 sizes.

Scale. The largest model is Qwen3-4B, we cover two model families, and we do not reproduce the LLaMA-2 results of the paper; effects that appear only at larger scale are outside this study. Most concepts use one prompt format, the template of the paper, and the wording study varies it for sentiment only.

Test size and intervals. Truth and honesty are scored on 59 test pairs and TruthfulQA on 173 questions, so differences below about 0.176 to 0.179 on GPT-2, 0.126 to 0.167 on Qwen3 base and 0.132 to 0.170 on Qwen3 post-trained checkpoints are not resolved; GPT-2 truth readings are inconclusive rather than negative. Layers are selected on small validation sets, so the chosen layer carries its own uncertainty. Our intervals cover the sampling of test items only: each cell is one run, so variation between runs, prompts or seeds of the fitting procedure is not included, and the difference intervals ignore the pairing of the arms and are conservative.

Base against post-trained. A post-trained Qwen3 checkpoint differs from its base in post-training data, chat format and distillation, so a difference is a contrast between those recipes. The stored per-seed comparison covers only the 0.6B and 1.7B pairs; Figure 10 adds the 4B pair from the run summaries.

Unmatched pairs. The finder of the paper is tested on matched pairs only for utility; for sentiment and truth our pairs are unrelated inputs, and we did not build matched pairs for them.

Judges. Sentiment is the positive-class probability of a classifier, the utility steering test is judged by the same sentiment classifier, which makes one judge family, and abstention is detected by phrase matching without human labels, which does not separate a generic answer to an unknowable detail from a specific invented one, because the runs do not label which a non-abstaining answer is.

Labels and approximate answers. Known and unknown questions are defined by the answers of each model, so question sets differ between models and an AUROC is not a measure of the same questions across models; the labels of knowledge questions are alias string matches with no hand check, and GPT-2 small has too few known questions for a contrast. For approximate answers the range side is a living thing and the exact side an artifact in every family, the wording differs by side, test sets hold 22 to 30 questions per side, and an item bootstrap interval collapses at an AUROC of 1.000.

Controls and comparisons. The readings are compared with model-free word and character n-gram classifiers from a separate run (Table 2), which see only the text, not the activations at the layer of the reader; the honesty baseline is fitted on the true and false labels of the facts, not on the persona labels, so it is a baseline for the transfer pairs only, and its interval covers test-pair sampling only. The sentiment steering run has no random-direction arm (the utility steering run has three), and knowledge steering has three random directions per run, so a single run passes against them by chance with probability 1/4. The page reports many comparisons across 10 checkpoints and counts how many pass; the counts are descriptive and not corrected for the number of comparisons.

6 Reproducibility

Every number on this page is computed when the page is built, from the result files in the results directory of the repository; each figure and table links the data file it draws, and the tests of the site recompute every number in a caption from the result files. Links to the findings document open it at the commit of these results. Each run directory holds its configuration, its run identity (code commit and configuration hash) and its outputs. Models and datasets are pinned to the revisions in Table A1. The experiments ran on a single consumer GPU, on rented single-GPU cloud machines and on a 16-thread CPU machine. One experiment trains a model beyond the readers: a LoRRA adapter fitted on all 10 checkpoints (Appendix J). Every experiment is one command, tomography run followed by the experiment and the checkpoint, with base seed 0 from which the seeds of the readers are derived by a hash, so a rerun reproduces its files; the section of the README on running experiments lists the command and the options of each (Table A2 explains the run codes of the run directories).

The code, the tests and every result are in the repository, the findings document (every result with its run identity), the methodology document (deviations from the paper and the statistics) and the committed result files. The page is rebuilt from those files by the commands below.

```
uv sync
uv run tomography run r3 --model gpt2
uv run python -m tomography.site build --out _site
```

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Cite this repository

This repository reimplements the method of the paper cited in the introduction; please cite the original work as well.

```
@misc{tomography,
  title = {Reading and Steering Concepts with One Direction: Representation Engineering on GPT-2 and Qwen3},
  howpublished = {\url{https://github.com/javierdejesusda/tomography}},
  year = {2026},
}

@misc{zou2025representationengineeringtopdownapproach,
      title={Representation Engineering: A Top-Down Approach to {AI} Transparency},
      author={Andy Zou and Long Phan and Sarah Chen and James Campbell and Phillip Guo and Richard Ren and Alexander Pan and Xuwang Yin and Mantas Mazeika and Ann-Kathrin Dombrowski and Shashwat Goel and Nathaniel Li and Michael J. Byun and Zifan Wang and Alex Mallen and Steven Basart and Sanmi Koyejo and Dawn Song and Matt Fredrikson and J. Zico Kolter and Dan Hendrycks},
      year={2023},
      eprint={2310.01405},
      archivePrefix={arXiv},
      primaryClass={cs.LG},
      url={https://arxiv.org/abs/2310.01405},
      note={arXiv:2310.01405v4, 2025},
}
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Appendix A Assets and notation

Table A1 lists every checkpoint, judge and dataset with its pinned revision and the license its source states, which is what makes the results reproducible, and Table A2 defines the symbols and terms of the paper.

Table A1: Models, judges and datasets with their pinned revisions and their licenses as stated by their sources. Revision is the leading 12 characters of the pinned commit of the Hugging Face Hub repository or Git repository, or the SHA-256 digest of the pinned file. Sources without a permissive license are downloaded at run time and are not redistributed in the repository.

Asset	Used for	Citation	Hub ID or source	Revision	License as stated
GPT-2 small	GPT-2 checkpoint	[3]	openai-community/gpt2	607a30d783df	Modified MIT
GPT-2 medium	GPT-2 checkpoint	[3]	openai-community/gpt2-medium	6dcaa7a952f7	Modified MIT
GPT-2 large	GPT-2 checkpoint	[3]	openai-community/gpt2-large	32b71b12589c	Modified MIT
GPT-2 XL	GPT-2 checkpoint	[3]	openai-community/gpt2-xl	15ea56dee5df	Modified MIT
Qwen3-0.6B-Base	Qwen3 checkpoint	[4]	Qwen/Qwen3-0.6B-Base	da87bfb608c1	Apache-2.0
Qwen3-0.6B	Qwen3 checkpoint	[4]	Qwen/Qwen3-0.6B	c1899de289a0	Apache-2.0
Qwen3-1.7B-Base	Qwen3 checkpoint	[4]	Qwen/Qwen3-1.7B-Base	ea980cb0a6c2	Apache-2.0
Qwen3-1.7B	Qwen3 checkpoint	[4]	Qwen/Qwen3-1.7B	70d244cc86cc	Apache-2.0
Qwen3-4B-Base	Qwen3 checkpoint	[4]	Qwen/Qwen3-4B-Base	906bfd4b4dc7	Apache-2.0

Asset	Used for	Citation	Hub ID or source	Revision	License as stated
Qwen3-4B	Qwen3 checkpoint	[4]	Qwen/Qwen3-4B	1cfa9a720891	Apache-2.0
SiEBERT	judge of the sentiment of generated text	[30]	siebert/sentiment-roberta-large-english	74cea614e245	None declared
DistilBERT SST-2	second judge of the sentiment of generated text	[34]	distilbert/distilbert-base-uncased-finetuned-sst-2-english	714eb0fa89d2	Apache-2.0
SST-2	sentiment pairs	[26]	stanfordnlp/sst2	8d51e7e4887a	Unknown (cite Socher et al., 2013 [26])
ETHICS	utility and morality pairs	[27]	people.eecs.berkeley.edu/~hendrycks/ethics.tar	SHA-256 40acbf1ac0da	MIT
True and false facts of the representation engineering repository	truth and honesty statements	[1] [10]	github.com/andyzoujm/representation-engineering	5455d8a375d5	MIT (granted by the representation-engineering repository)
Geometry of Truth	out-of-distribution truth sets	[11]	github.com/saprmarks/geometry-of-truth	5d1c630c44f7	None stated (runtime download only)
CommonClaim	common claims set, as shipped with Geometry of Truth	[35]	github.com/saprmarks/geometry-of-truth	5d1c630c44f7	None stated (runtime download only)
TruthfulQA	multiple-choice truthfulness questions	[28]	truthfulqa/truthful_qa	741b8276f2d1	Apache-2.0
ARC	multiple-choice questions for a second reader source	[36]	allenai/ai2_arc	210d026faf99	CC-BY-SA-4.0
PopQA	known and unknown questions, subject popularity	[23]	akariasai/PopQA	098765c79ea1	No license on the dataset card (code repository: MIT)
TriviaQA	known and unknown questions	[37]	mandarjoshi/trivia_qa	0f7faf33a390	Unknown on the dataset card, Apache-2.0 on the GitHub repository
KUQ	known and unknown questions	[38]	amayuelas/KUQ	f99b53aa226d	MIT
FalseQA	questions with false premises	[39]	github.com/thunlp/FalseQA	9a9729974d8c	No license: runtime download only
SyntHal	synthetic unanswerable questions	[40]	github.com/yuetianluNLP/SyntHal	c4fe4cecf41b	Research and review terms, no permissive license
NQ-open	known and unknown questions	[41] [42]	google-research-datasets/nq_open	5dd9790a8300	CC-BY-SA-3.0
Wikidata	quantities and award questions	[29]	query.wikidata.org/sparql	snapshot date and SHA-256 per item	CC0

Asset	Used for	Citation	Hub ID or source	Revision	License as stated
WikiText-103	held-out passages for perplexity	[43]	Salesforce/wikitext	b08601e04326	CC-BY-SA-3.0 and GFDL

Table A2: Symbols and terms used in the text. Experiment names in the repository are given in the last row.

Symbol or term	Meaning
pair	Two inputs that differ in the concept: a positive member (label 1) and a negative member (label 0).
h(x, ℓ)	Residual-stream activation of the last token of input x after layer ℓ.
v(ℓ)	Unit direction at layer ℓ found by a finder on the training pairs and oriented with their labels.
u(ℓ)	Random unit direction at layer ℓ, oriented on the same training pairs.
finder	Method that turns training pairs into a direction: PCA of paired differences (the paper's), PCA of activations, mean difference, logistic regression or k-means.
accuracy	Share of held-out pairs whose positive member projects higher on the direction; ties count one half; 0.5 for a direction with no information.
margin	Accuracy of v minus accuracy of u on the same test pairs, both at the same layer except in the finder comparison, which reads each at the layer chosen on its own select split.
relative depth	Layer index divided by the number of layers.
layer chosen on validation	Layer with the highest accuracy on validation or select pairs that the reported score does not use.
AUROC	Probability that a positive item scores above a negative one; 0.5 is chance.
relative coefficient c	Steering strength as a multiple of r, the mean residual norm at layer ℓ on calibration text.
win rate	Share of decisive prompt pairs in which the completion steered at +c is judged more positive than the one steered at −c.
role	GPT-2, Qwen3 base or Qwen3 post-trained; every count is given per role.
run codes	The directories of the results folder are named by an experiment code, a letter and a number; the page names experiments by what they test and the README lists the codes.

Appendix B Method details

These are the design details that the method summarises; they support every result of the paper, because each one is reported against a baseline built the same way.

Readers and draws. A run fits 10 readers (15 for TruthfulQA and 20 for the held-out sentiment run), each on its own draw of 64 training pairs, and reports the mean of their accuracies. The honesty readers read the response tokens of the persona prompts instead of the last token of the input.

Layer choice. The layer is picked on a validation split for utility, morality and truth, on 53 validation statements for the honesty transfer, on a select split for the finder comparison and the sample-size run, and on validation questions for TruthfulQA. The main sentiment run chooses its layer on the 256 SST-2 validation pairs it is scored on; the sample-size run scores a separate report split of 214 pairs.

Data and judges. Truth is read on the true and false statements that the representation engineering repository [1] derives from Azaria and Mitchell [10], and tested out of distribution on the Geometry of Truth sets [11], whose common claims come from Casper

et al. [35]; honesty is read on the same statements under the persona prompts of the paper. TruthfulQA [28] and ARC [36] give multiple-choice questions. Knowledge is read on questions the model can or cannot answer, drawn from PopQA [23], TriviaQA [37], KUQ [38], FalseQA [39], SyntHal [40], NQ-open [41] [42] and Wikidata [29]. Questions are labelled known or unknown from the behaviour of each model, so each model has its own question sets. Generated text is scored by SiEBERT, with an SST-2 DistilBERT classifier [34] as a check; perplexity is measured on held-out WikiText-103 passages [43].

Baselines. The popularity of the subject is its monthly Wikipedia page views [23]. The confidence of the model stacks five output signals, and only the pooled level of the stack is stored.

Steering. The sentiment steering run uses the logistic sentiment direction on all 10 checkpoints, with 3 steering vectors and the relative coefficients −0.3 to 0.3, and compares steering with a positive or negative instruction. Steered text is fluent only when the perplexity of 200 held-out WikiText-103 passages under the steered model is at most 1.5 times that under the unsteered model and the distinct-1 of the completions is at least 0.15. Table A10 also gives the perplexity of the completions themselves under the unsteered model. The steering comparison of finders follows the manipulation test of the paper on ETHICS utility: for the same prompt and seed, the completion steered at +0.3 is compared with the one at −0.3, the sentiment judge decides which is more positive, and the win rate is the share of decisive pairs won by the +0.3 completion. Knowledge steering adds the known-against-unknown direction while the model answers; base checkpoints cannot abstain, so their outcome is the shift of the entropy of the next-token distribution, and post-trained checkpoints are asked to say when they do not know, so their outcome is the abstention rate, detected by phrase matching.

Intervals. The intervals stored by the reading runs are percentile bootstrap intervals over the 10 readers; they do not resample the test pairs and are narrower than the sampling of the test pairs. The interval for an accuracy on n test pairs is Wilson, and for a difference between two accuracies it is d ± 1.96 times the square root of p1 (1 − p1) / n1 + p2 (1 − p2) / n2, which treats the two arms as independent and is conservative because they share the test items. AUROC intervals use the numbers of positive and negative test items; increments in error prediction keep their stored paired bootstrap over test questions; win rates keep their Wilson interval on decisive pairs; slopes over the four GPT-2 sizes are least-squares slopes with a t interval on 2 degrees of freedom.

Appendix C Reading details

These support the finding that a logistic direction reads utility and morality on every checkpoint and truth on Qwen3 only: the margin by depth and role (Figure A1), the layer curves of every reading (Figure A2), every finder (Figure A3, Figure A4, Table A3), every reading at its layer (Table A4) and the sample-size and decoy runs (Table A5, Table A6).

Baselines of each row of Figure 1: rows 1 to 5, the random direction of the same run at the same layer; row 6, the random direction of the finder comparison; row 7, a word

classifier; row 8, the strongest of popularity and two text classifiers; row 9, the model's own confidence, five stacked output signals. A direction for questions the model can answer is above that confidence, with an interval excluding zero, on 0 of 3 GPT-2, 0 of 3 Qwen3 base and 0 of 3 post-trained checkpoints. For steering on the utility manipulation task of the paper, a checkpoint passes when every logistic steering vector (3 per checkpoint) of the utility direction beats every random direction of the same norm (3 per checkpoint) on the sentiment judge at relative coefficient 0.3, which has chance 0.05: 1 of 4 GPT-2, 2 of 3 Qwen3 base and 2 of 3 post-trained checkpoints. At that coefficient the steered text meets the fluency rule (perplexity at most 1.5 times the unsteered text and distinct-1 at least 0.15) on 0 of 3 GPT-2, 1 of 3 Qwen3 base and 0 of 3 post-trained checkpoints (GPT-2 small has no fluency measure). Intervals cover the sampling of the test items and treat the two arms as independent, which makes them conservative; rows 8 and 9 use the Hanley and McNeil approximation and the stored paired bootstrap.

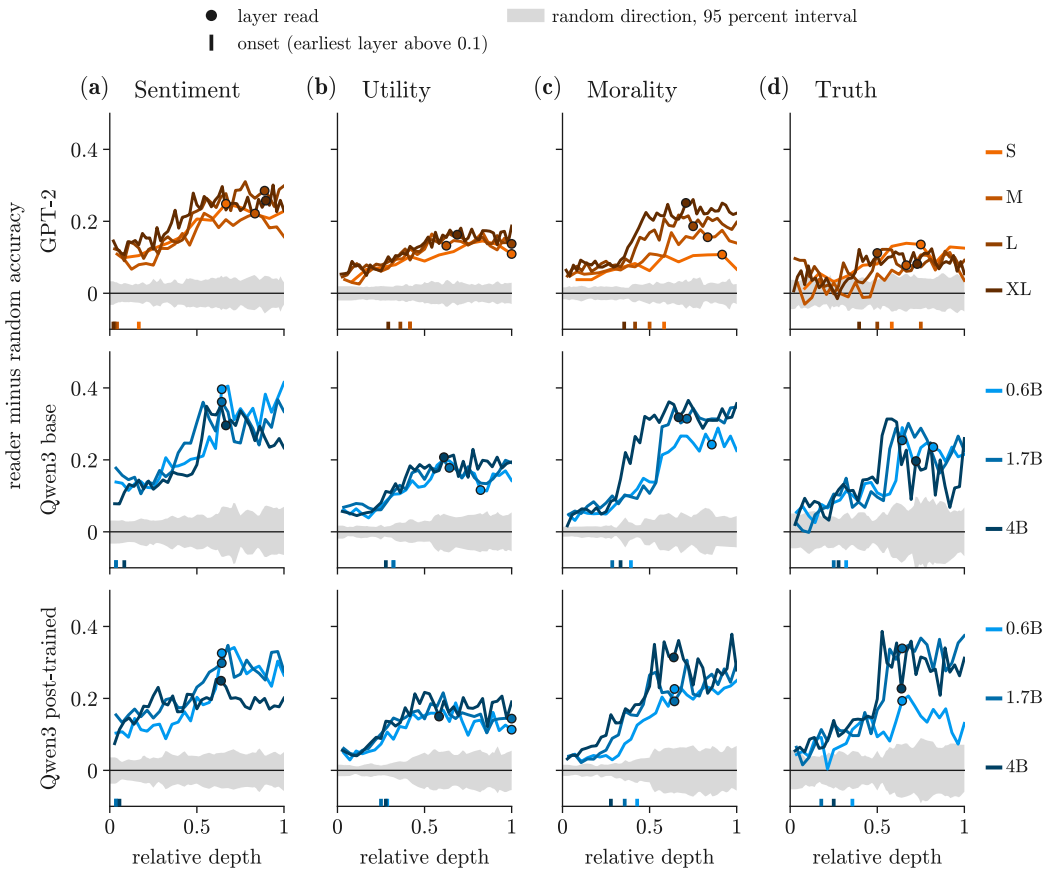


Figure A1: A reader's margin over a random direction reaches 0.1 at a relative depth of sentiment 0.021 to 0.167 on GPT-2, 0.036 to 0.083 on Qwen3 base and 0.036 to 0.056 on post-trained; utility 0.292 to 0.417 on GPT-2, 0.278 to 0.321 on Qwen3 base and 0.250 to 0.286 on post-trained; morality 0.354 to 0.583 on GPT-2, 0.286 to 0.393 on Qwen3 base and 0.278 to 0.429 on post-trained; truth 0.396 to 0.750 on GPT-2, 0.250 to 0.321 on Qwen3 base and 0.179 to 0.357 on post-trained. Reader accuracy minus random-direction accuracy at every layer, one line per checkpoint in the panel of its role (rows: GPT-2, Qwen3 base, Qwen3 post-trained; darker lines are larger checkpoints, named in the right column of each row). The dot marks the layer read, chosen on a validation split (for sentiment the layer of the finder comparison); the tick on the x axis marks the onset, the earliest layer at which the margin curve crosses 0.1, which is not an estimate with an interval; for truth the curve rests on 59 truth pairs. The grey band is the stored 95 percent interval over seeds of the random direction, drawn around the zero line and spanning the checkpoints of the panel. Sentiment curves are the report-split accuracy of the logistic regression of the finder comparison. The accuracy at the layer read against the checkpoint size is in Figure 4.

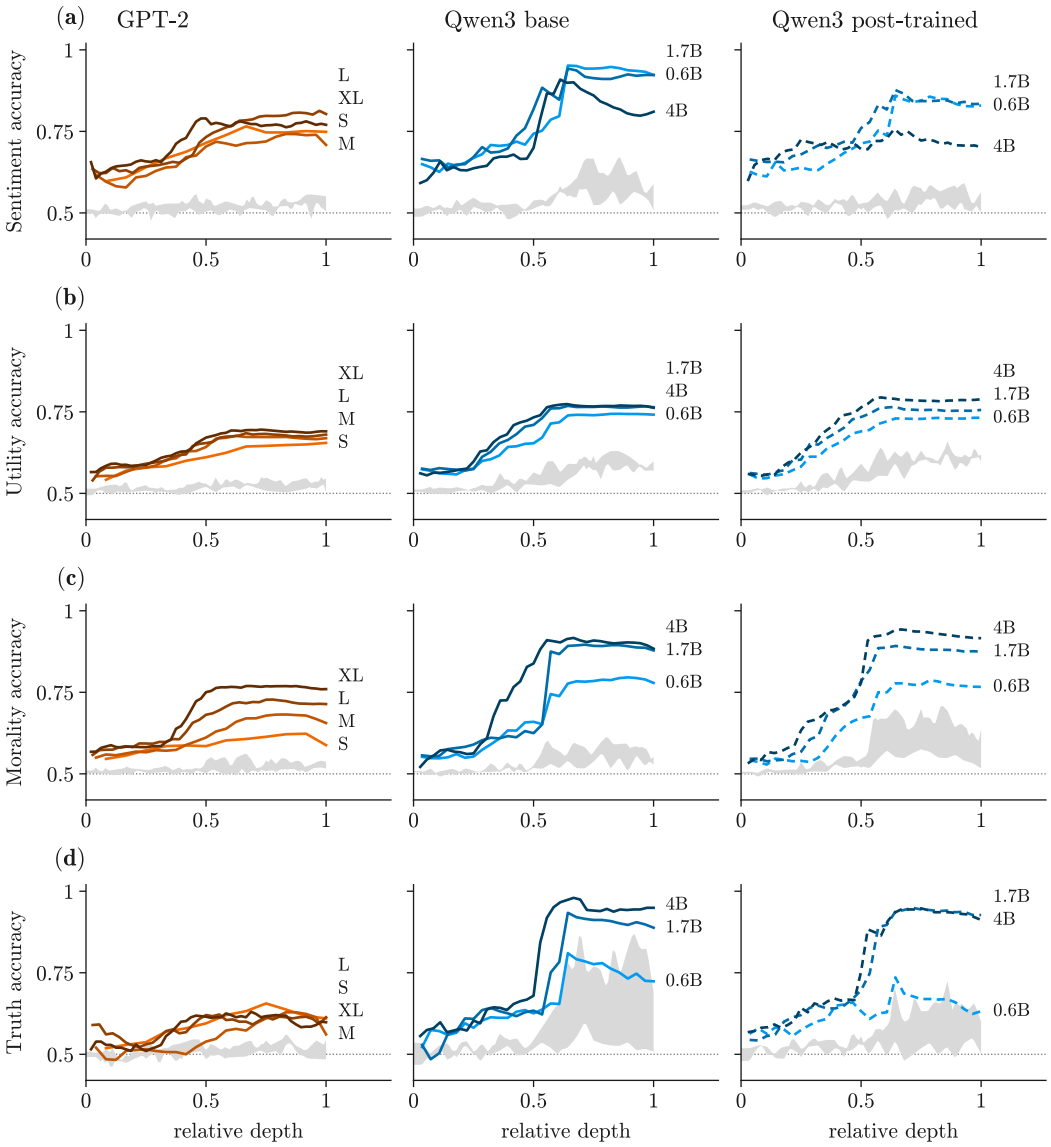


Figure A2: Reader accuracy at every layer for every concept and checkpoint, against the range of the random directions of the same panel (grey). Sentiment curves are the report-split accuracy of the logistic regression of the finder comparison on 214 sentiment pairs. Lines are the mean over seeds of a logistic regression reader; colour darkens with model size within a family, post-trained checkpoints are dashed, labels at the right name the size, and the dotted line is chance.

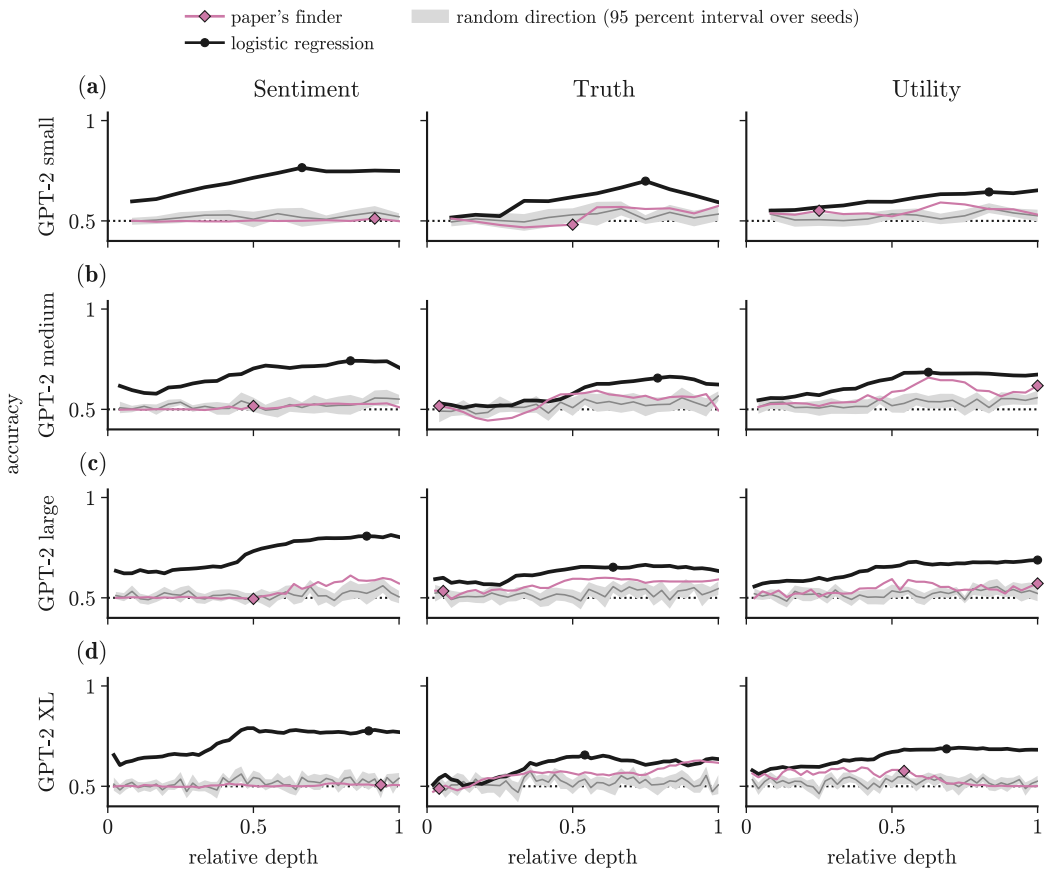


Figure A3: Report-split accuracy at every layer of the paper's finder (purple) and logistic regression (black) on GPT-2, against the random direction of the same run (grey). Dots mark the layer each finder's select split chose. The grey band is the 95 percent interval over seeds of the random direction, with its mean as a thin line, and the dotted line is chance. The other finders are in Table A3.

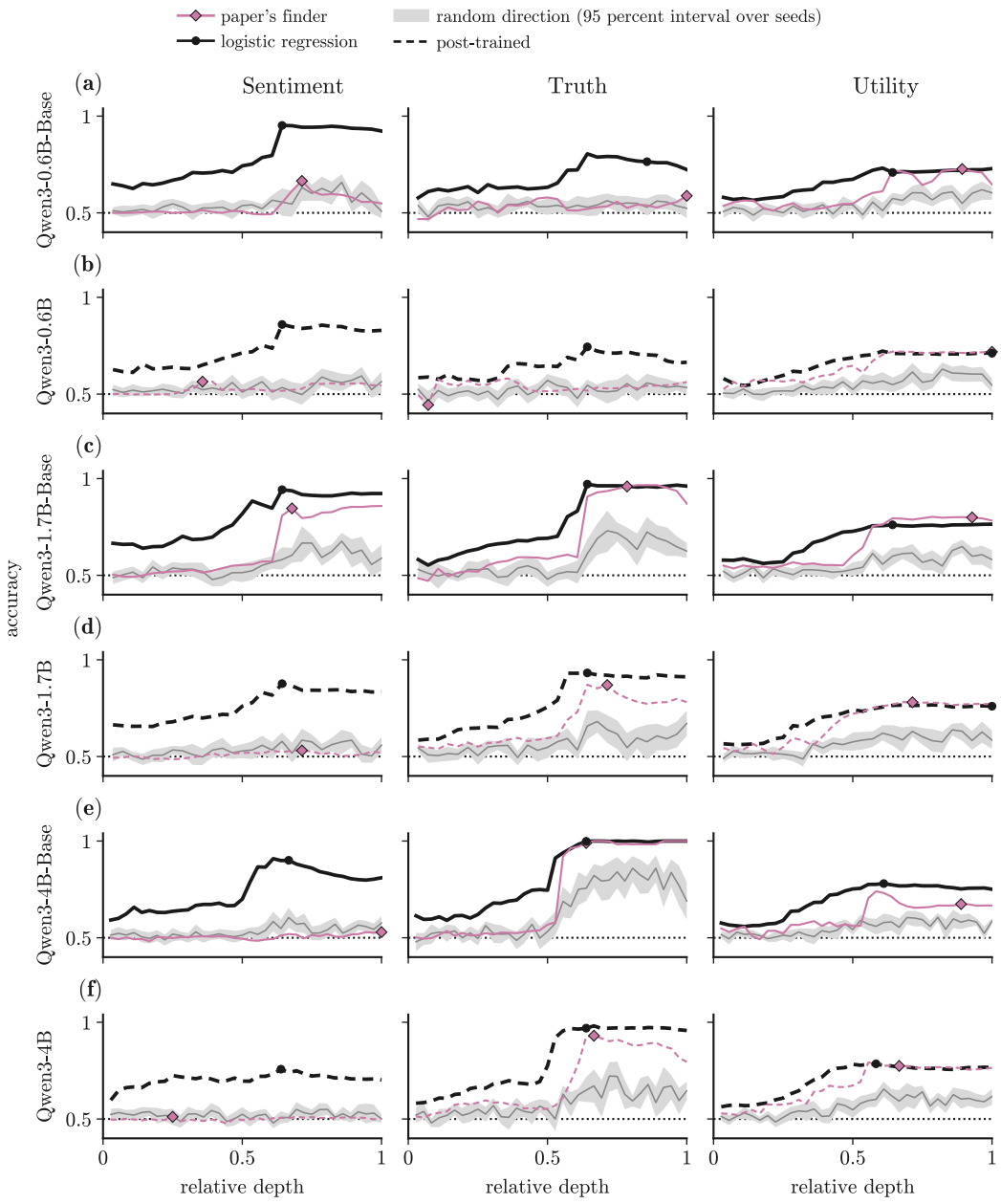


Figure A4: Report-split accuracy at every layer of the paper's finder (purple) and logistic regression (black) on Qwen3, against the random direction of the same run (grey). Dots mark the layer each finder's select split chose. The grey band is the 95 percent interval over seeds of the random direction, with its mean as a thin line, and the dotted line is chance. Post-trained checkpoints are dashed. The other finders are in Table A3.

Table A3: Every finder on the report split, as accuracy minus the random direction (random row: its accuracy), and the manipulation win rate of directions fitted on utility pairs. Reading cells use the layer each finder chose on its select split and an asterisk marks a conservative 95 percent interval for the report pairs that excludes zero. The win rate is the share of decisive prompt pairs in which the completion steered at $+0.3$ is judged more positive than the one steered at -0.3 by the sentiment judge, with the Wilson half-width. Three random directions of the same norm are the reference arm for the win rate; 0.5 is not.

Group	Finder	GPT-2 small	GPT-2 medium	GPT-2 large	GPT-2 XL	Qwen3-0.6B-Base	Qwen3-0.6B	Qwen3-1.7B-Base	Qwen3-1.7B	Qwen3-4B-Base	Qwen3-4B
Sentiment	PCA of paired differences (paper)	−0.031	−0.040	−0.030	−0.038	0.007	−0.029	0.170*	−0.053	−0.047	−0.014
	PCA of activations	−0.047	−0.047	−0.024	−0.039	−0.101*	−0.020	0.161*	−0.063	−0.037	−0.020
	mean difference	0.124*	0.133*	0.194*	0.118*	0.279*	0.156*	0.217*	0.166*	0.215*	0.090
	logistic regression	0.222*	0.186*	0.283*	0.231*	0.294*	0.266*	0.266*	0.292*	0.324*	0.233*
	k-means	−0.048	−0.050	−0.023	−0.045	−0.082	−0.043	0.171*	−0.062	−0.049	−0.015
	random direction	0.543	0.557	0.525	0.545	0.658	0.594	0.676	0.585	0.577	0.525
Truth	PCA of paired differences (paper)	−0.029	−0.019	0.000	−0.015	0.039	−0.114	0.236*	0.231*	0.131*	0.208*
	PCA of activations	−0.036	−0.063	0.007	−0.017	−0.002	−0.124	0.220*	0.142	0.120*	0.166*
	mean difference	0.027	0.080	0.012	0.010	0.139	0.042	0.239*	0.273*	0.136*	0.219*
	logistic regression	0.188*	0.120	0.119	0.153	0.215*	0.186*	0.247*	0.293*	0.139*	0.247*
	k-means	−0.032	−0.022	0.005	−0.024	−0.017	−0.108	0.227*	0.169*	0.136*	0.212*
	random direction	0.510	0.536	0.534	0.503	0.549	0.558	0.724	0.639	0.859	0.722
Utility	PCA of paired differences (paper)	0.025	0.065	0.026	0.051	0.122*	0.104*	0.150*	0.154*	0.093*	0.120*
	PCA of activations	0.029	0.083	0.065	0.071	0.133*	0.105*	0.144*	0.148*	0.143*	0.121*
	mean difference	0.101*	0.100*	0.112*	0.115*	0.130*	0.106*	0.142*	0.157*	0.205*	0.127*
	logistic regression	0.118*	0.132*	0.143*	0.161*	0.105*	0.097*	0.112*	0.133*	0.199*	0.130*
	k-means	0.032	0.080	0.068	0.057	0.136*	0.098*	0.139*	0.152*	0.164*	0.120*
	random direction	0.526	0.553	0.546	0.525	0.604	0.614	0.649	0.627	0.581	0.655
Win rate at 0.3 (utility directions, sentiment judge)	PCA of paired differences (paper)	0.713 ± 0.032	0.728 ± 0.032	0.563 ± 0.035	0.755 ± 0.031	0.715 ± 0.032	0.913 ± 0.020	0.805 ± 0.028	0.920 ± 0.019	0.911 ± 0.020	0.951 ± 0.016
	PCA of activations	0.672 ± 0.034	0.671 ± 0.034	0.537 ± 0.036	0.716 ± 0.032	0.508 ± 0.036	0.927 ± 0.019	0.757 ± 0.031	0.919 ± 0.020	0.925 ± 0.019	0.965 ± 0.013
	mean difference	0.903 ± 0.021	0.659 ± 0.034	0.600 ± 0.035	0.727 ± 0.032	0.711 ± 0.032	0.907 ± 0.021	0.896 ± 0.022	0.920 ± 0.019	0.949 ± 0.016	0.969 ± 0.013
	logistic regression	0.925 ± 0.019	0.495 ± 0.036	0.540 ± 0.036	0.649 ± 0.034	0.743 ± 0.031	0.772 ± 0.030	0.807 ± 0.028	0.743 ± 0.031	0.869 ± 0.024	0.855 ± 0.025
	k-means	0.592 ± 0.035	0.728 ± 0.032	0.533 ± 0.036	0.611 ± 0.035	0.524 ± 0.036	0.919 ± 0.020	0.752 ± 0.031	0.921 ± 0.019	0.937 ± 0.017	0.960 ± 0.014
	random direction	0.432 ± 0.035	0.547 ± 0.036	0.441 ± 0.035	0.492 ± 0.036	0.477 ± 0.036	0.549 ± 0.036	0.480 ± 0.036	0.477 ± 0.036	0.529 ± 0.036	0.460 ± 0.036

Table A4: Every reading at its reported layer, with its margin over the random direction. The sentiment block without a held-out split chooses the layer on the same 256 SST-2 validation pairs it is scored on and is optimistic; the held-out block chooses it on the select split of the sample-size run and scores 214 pairs. Margins carry conservative 95 percent intervals for the test pairs; the intervals stored with the runs resample the readers only and are in the data files. Honesty in template is scored on held-out persona prompts and honesty transfer on plain true and false statements.

Concept	Model	Layer	Relative depth	Reader	Random	Margin	Test pairs
Sentiment, layer chosen on the scored pairs (optimistic)	GPT-2 small	8	0.667	0.772	0.518	0.254 ± 0.080	256
	GPT-2 medium	20	0.833	0.789	0.549	0.240 ± 0.079	256
	GPT-2 large	35	0.972	0.838	0.518	0.320 ± 0.076	256

Concept	Model	Layer	Relative depth	Reader	Random	Margin	Test pairs
	GPT-2 XL	24	0.500	0.817	0.523	0.293 ± 0.077	256
	Qwen3-0.6B-Base	19	0.679	0.947	0.602	0.345 ± 0.066	256
	Qwen3-0.6B	18	0.643	0.857	0.544	0.313 ± 0.075	256
	Qwen3-1.7B-Base	18	0.643	0.905	0.620	0.286 ± 0.069	256
	Qwen3-1.7B	18	0.643	0.866	0.557	0.309 ± 0.074	256
	Qwen3-4B-Base	22	0.611	0.905	0.543	0.363 ± 0.071	256
	Qwen3-4B	23	0.639	0.760	0.507	0.252 ± 0.081	256
Sentiment, held out	GPT-2 small	8	0.667	0.758	0.537	0.221 ± 0.088	214
	GPT-2 medium	15	0.625	0.754	0.518	0.236 ± 0.088	214
	GPT-2 large	24	0.667	0.805	0.542	0.263 ± 0.085	214
	GPT-2 XL	26	0.542	0.800	0.533	0.267 ± 0.086	214
	Qwen3-0.6B-Base	28	1.000	0.919	0.547	0.372 ± 0.076	214
	Qwen3-0.6B	26	0.929	0.834	0.565	0.269 ± 0.083	214
	Qwen3-1.7B-Base	18	0.643	0.937	0.602	0.336 ± 0.073	214
	Qwen3-1.7B	18	0.643	0.888	0.574	0.314 ± 0.079	214
	Qwen3-4B-Base	23	0.639	0.913	0.570	0.343 ± 0.076	214
	Qwen3-4B	23	0.639	0.781	0.523	0.258 ± 0.087	214
Utility	GPT-2 small	12	1.000	0.655	0.546	0.109 ± 0.019	4,808
	GPT-2 medium	15	0.625	0.676	0.544	0.132 ± 0.019	4,808
	GPT-2 large	36	1.000	0.680	0.543	0.137 ± 0.019	4,808
	GPT-2 XL	33	0.688	0.694	0.532	0.163 ± 0.019	4,808
	Qwen3-0.6B-Base	23	0.821	0.744	0.628	0.116 ± 0.018	4,808
	Qwen3-0.6B	28	1.000	0.732	0.619	0.113 ± 0.019	4,808
	Qwen3-1.7B-Base	18	0.643	0.768	0.591	0.178 ± 0.018	4,808

Concept	Model	Layer	Relative depth	Reader	Random	Margin	Test pairs
	Qwen3-1.7B	28	1.000	0.756	0.612	0.144 ± 0.018	4,808
	Qwen3-4B-Base	22	0.611	0.772	0.565	0.207 ± 0.018	4,808
	Qwen3-4B	21	0.583	0.794	0.645	0.150 ± 0.018	4,808
Morality	GPT-2 small	11	0.917	0.623	0.516	0.108 ± 0.044	983
	GPT-2 medium	20	0.833	0.682	0.526	0.156 ± 0.043	983
	GPT-2 large	27	0.750	0.727	0.541	0.186 ± 0.042	983
	GPT-2 XL	34	0.708	0.768	0.517	0.251 ± 0.041	983
	Qwen3-0.6B-Base	24	0.857	0.793	0.551	0.242 ± 0.040	983
	Qwen3-0.6B	18	0.643	0.778	0.551	0.226 ± 0.041	983
	Qwen3-1.7B-Base	20	0.714	0.896	0.582	0.314 ± 0.036	983
	Qwen3-1.7B	18	0.643	0.892	0.700	0.192 ± 0.035	983
	Qwen3-4B-Base	24	0.667	0.917	0.598	0.318 ± 0.035	983
	Qwen3-4B	23	0.639	0.940	0.626	0.314 ± 0.034	983
Truth	GPT-2 small	9	0.750	0.656	0.520	0.136 ± 0.176	59
	GPT-2 medium	16	0.667	0.573	0.495	0.078 ± 0.179	59
	GPT-2 large	18	0.500	0.620	0.508	0.112 ± 0.178	59
	GPT-2 XL	35	0.729	0.614	0.532	0.081 ± 0.178	59
	Qwen3-0.6B-Base	23	0.821	0.763	0.527	0.236 ± 0.167	59
	Qwen3-0.6B	18	0.643	0.736	0.542	0.193 ± 0.170	59
	Qwen3-1.7B-Base	18	0.643	0.934	0.680	0.254 ± 0.135	59
	Qwen3-1.7B	18	0.643	0.937	0.598	0.339 ± 0.140	59
	Qwen3-4B-Base	26	0.722	0.942	0.746	0.197 ± 0.126	59
	Qwen3-4B	23	0.639	0.934	0.707	0.227 ± 0.132	59
Honesty in template	GPT-2 small	1	0.083	1.000	0.832	0.168 ± 0.086	72

Concept	Model	Layer	Relative depth	Reader	Random	Margin	Test pairs
	GPT-2 medium	1	0.042	1.000	0.808	0.192 ± 0.091	72
	GPT-2 large	1	0.028	1.000	0.763	0.237 ± 0.098	72
	GPT-2 XL	1	0.021	1.000	0.714	0.286 ± 0.104	72
	Qwen3-0.6B-Base	1	0.036	1.000	0.617	0.383 ± 0.112	72
	Qwen3-0.6B	1	0.036	1.000	0.636	0.364 ± 0.111	72
	Qwen3-1.7B-Base	1	0.036	1.000	0.609	0.391 ± 0.113	72
	Qwen3-1.7B	1	0.036	1.000	0.610	0.390 ± 0.113	72
	Qwen3-4B-Base	2	0.056	1.000	0.606	0.394 ± 0.113	72
	Qwen3-4B	1	0.028	1.000	0.657	0.343 ± 0.110	72
Honesty transfer	GPT-2 small	7	0.583	0.386	0.483	−0.097 ± 0.178	59
	GPT-2 medium	9	0.375	0.598	0.476	0.122 ± 0.179	59
	GPT-2 large	16	0.444	0.566	0.473	0.093 ± 0.180	59
	GPT-2 XL	1	0.021	0.492	0.502	−0.010 ± 0.180	59
	Qwen3-0.6B-Base	27	0.964	0.456	0.554	−0.098 ± 0.180	59
	Qwen3-0.6B	21	0.750	0.529	0.512	0.017 ± 0.180	59
	Qwen3-1.7B-Base	20	0.714	0.778	0.492	0.286 ± 0.166	59
	Qwen3-1.7B	20	0.714	0.781	0.485	0.297 ± 0.165	59
	Qwen3-4B-Base	36	1.000	0.658	0.531	0.127 ± 0.176	59
	Qwen3-4B	30	0.833	0.758	0.505	0.253 ± 0.168	59

Table A5: Held-out sentiment accuracy against the number of training pairs, with the random direction at the same layer in parentheses. Each cell is the mean over 20 draws of training pairs on 214 report pairs, at the layer the run reports.

Model	2	8	32	64	128	512
GPT-2 small	0.526 (0.509)	0.583 (0.506)	0.704 (0.521)	0.758 (0.537)	0.790 (0.511)	0.844 (0.527)
GPT-2 medium	0.523 (0.482)	0.588 (0.512)	0.684 (0.504)	0.754 (0.518)	0.797 (0.521)	0.869 (0.536)
GPT-2 large	0.514 (0.519)	0.593 (0.517)	0.721 (0.528)	0.805 (0.542)	0.855 (0.516)	0.899 (0.525)
GPT-2 XL	0.533 (0.489)	0.583 (0.512)	0.728 (0.545)	0.800 (0.533)	0.853 (0.545)	0.931 (0.527)
Qwen3-0.6B-Base	0.653 (0.511)	0.790 (0.574)	0.903 (0.538)	0.919 (0.547)	0.935 (0.578)	0.944 (0.565)

Model	2	8	32	64	128	512
Qwen3-0.6B	0.593 (0.524)	0.666 (0.543)	0.793 (0.538)	0.834 (0.565)	0.868 (0.565)	0.913 (0.555)
Qwen3-1.7B-Base	0.755 (0.506)	0.855 (0.512)	0.926 (0.557)	0.937 (0.602)	0.958 (0.575)	0.971 (0.585)
Qwen3-1.7B	0.586 (0.495)	0.692 (0.571)	0.840 (0.541)	0.888 (0.574)	0.907 (0.568)	0.946 (0.562)
Qwen3-4B-Base	0.590 (0.543)	0.696 (0.529)	0.860 (0.496)	0.913 (0.570)	0.942 (0.571)	0.962 (0.548)
Qwen3-4B	0.535 (0.509)	0.579 (0.516)	0.685 (0.532)	0.781 (0.523)	0.843 (0.517)	0.939 (0.528)

Table A6: Sentiment reading when a sentence stating a low or a high amount of the concept is added to the prompt. The sentence is drawn at random (the paper's decoy) or always contradicting the label. The reader is trained on the original prompts and read at the last token; each cell is the accuracy on 214 report pairs with its 95 percent Wilson half-width. It is a within-run contrast across prompt variants on the sentiment pairs of these runs, not the held-out split of Table 2; no random direction is stored for these runs, so the only reference is chance, 0.500. The random directions of Table A4 belong to other runs with their own training sample and layer, so they are not a baseline for these accuracies.

Model	Original	Random decoy	Against decoy
GPT-2 small	0.765 ± 0.056	0.763 ± 0.057	0.761 ± 0.057
GPT-2 medium	0.742 ± 0.058	0.724 ± 0.060	0.708 ± 0.060
GPT-2 large	0.807 ± 0.053	0.782 ± 0.055	0.749 ± 0.058
GPT-2 XL	0.776 ± 0.056	0.732 ± 0.059	0.714 ± 0.060
Qwen3-0.6B-Base	0.952 ± 0.030	0.790 ± 0.054	0.437 ± 0.066
Qwen3-0.6B	0.860 ± 0.047	0.837 ± 0.049	0.754 ± 0.057
Qwen3-1.7B-Base	0.944 ± 0.031	0.908 ± 0.039	0.785 ± 0.055
Qwen3-1.7B	0.880 ± 0.044	0.827 ± 0.051	0.722 ± 0.060
Qwen3-4B-Base	0.901 ± 0.040	0.958 ± 0.028	0.957 ± 0.028
Qwen3-4B	0.753 ± 0.057	0.754 ± 0.057	0.710 ± 0.060

Appendix D Wording

These support the finding that a change of wording moves a sentiment reader by more than the noise of the test pairs on 2 of 4 GPT-2, 3 of 3 Qwen3 base and 3 of 3 post-trained checkpoints. The span between the best and the worst wording, the bare reader against its random direction, the cosines, the rank of the paper wording and the noise are given per role in Figure A5; the transfer between wordings is in Figure A6 and the wordings themselves in Table A7.

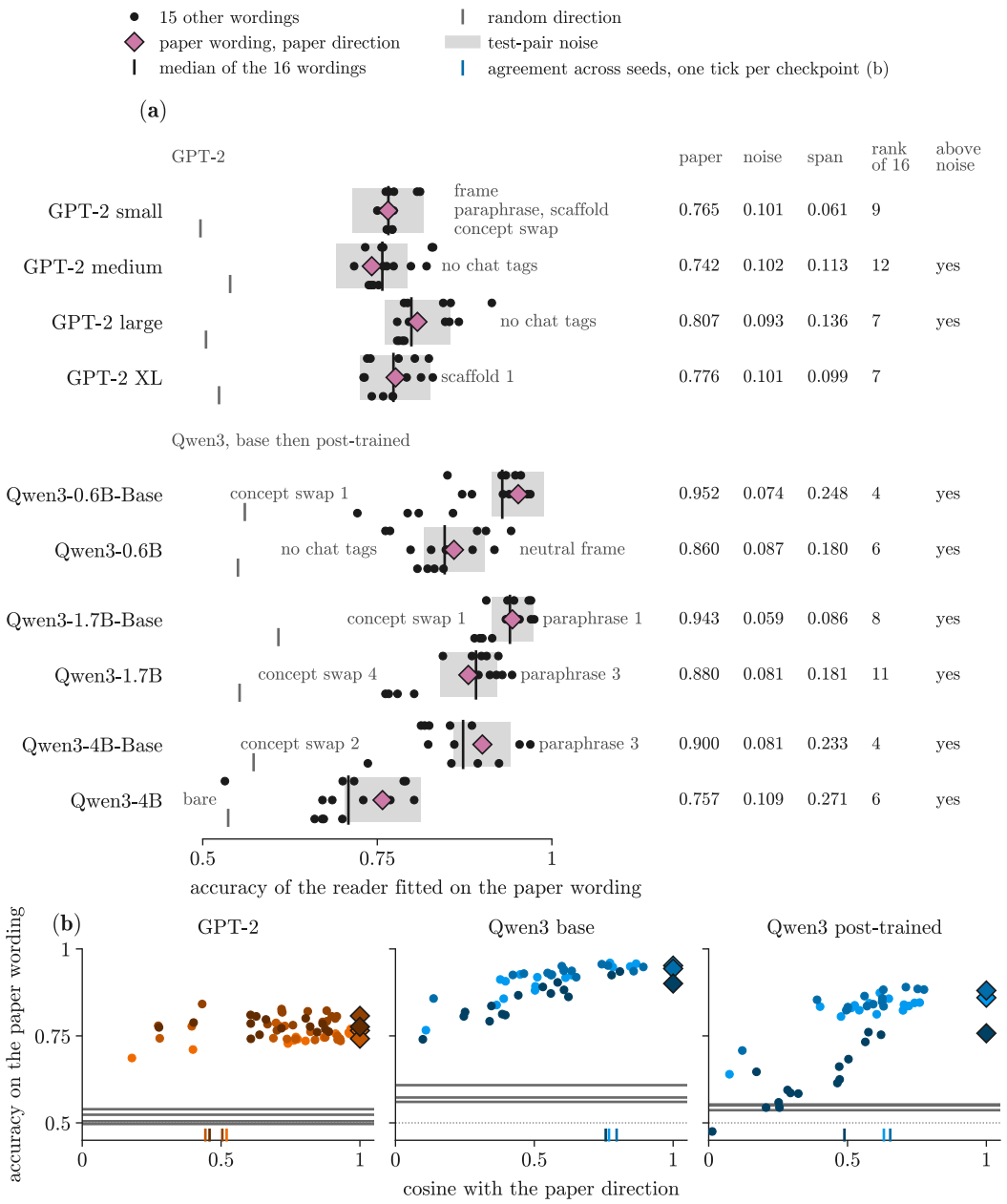


Figure A5: Wording moves a reader by more than the noise of the test pairs on 2 of 4 GPT-2, 3 of 3 Qwen3 base and 3 of 3 post-trained checkpoints (a point comparison of the span between the best and the worst wording with the noise expected from test pairs alone), and a direction at a large angle from the paper direction mostly still reads the paper wording above a random direction. The paper wording ranks 7 to 12 on GPT-2, 4 to 8 on Qwen3 base and 6 to 11 on post-trained of the 16 wordings by accuracy. A direction fitted on the bare wording has cosine 0.179 to 0.279 on GPT-2, 0.099 to 0.137 on Qwen3 base and 0.013 to 0.121 on post-trained with the paper direction. It reads the paper wording above a random direction by point estimate on 4 of 4 GPT-2, 3 of 3 Qwen3 base and 2 of 3 post-trained checkpoints; on Qwen3-4B the bare reader reads 0.475 against 0.536 for its random direction; it reads at 0.686 to 0.779 on GPT-2, 0.740 to 0.857 on Qwen3 base and 0.475 to 0.708 on post-trained, against 0.497 to 0.539 on GPT-2, 0.560 to 0.608 on Qwen3 base and 0.536 to 0.553 on post-trained for the random direction. (a) One row per checkpoint, base and post-trained checkpoints kept apart. Accuracy of the sentiment reader fitted on the paper's wording when it is

scored on each of the 15 other wordings: the 15 circles of a row sit in three lanes by kind of change (frame change, paraphrase or scaffold, concept swap, top to bottom, named in the first row); the diamond is the paper wording; the black tick is the median of the 16 wordings; the thin grey tick is the random direction; the best and the worst wording are named on the row unless they fall inside the grey band, with names as in Table A7. The band is centred on the paper wording and its full width is the noise expected from test pairs alone; it is conservative, because the wordings share the same test pairs, and it excludes fitting noise. The right columns give the paper accuracy, the noise, the span between the best and the worst wording, the rank of the paper wording among the 16 and a mark where the span exceeds the noise; the span is 0.061 to 0.136 on GPT-2, 0.086 to 0.248 on Qwen3 base and 0.180 to 0.271 on post-trained checkpoints, against 0.093 to 0.102 on GPT-2, 0.059 to 0.081 on Qwen3 base and 0.081 to 0.109 on post-trained expected from test-pair noise alone. (b) For each role, each direction fitted on another wording, by its cosine with the paper direction and its accuracy on the paper wording; the diamond at cosine 1 is the paper direction itself, ticks on the x axis mark how well the paper direction agrees with itself across seeds, and thin grey lines the random direction on the paper wording; colour darkens with size within a family. Matrices for every pair of wordings are in Figure A6.

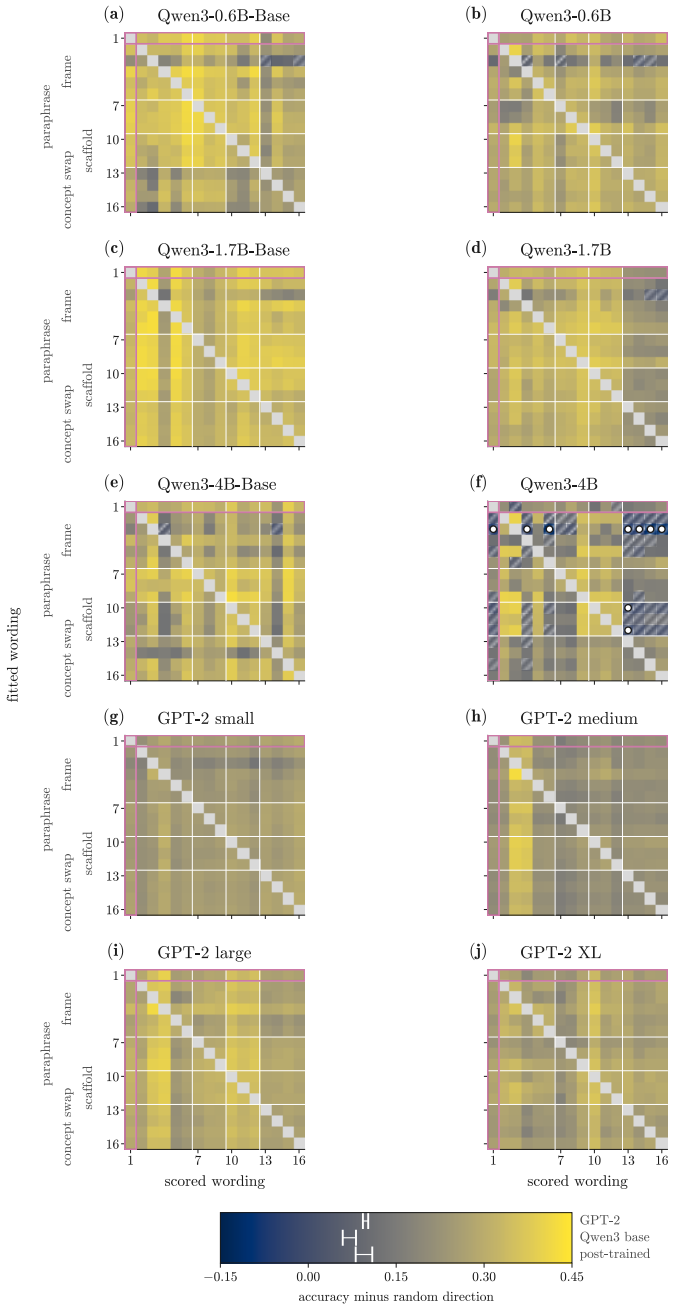


Figure A6: Accuracy of a reader fitted on one wording (row) and scored on another (column), minus the random direction on the scored wording, for every checkpoint with a wording run. The margin is at or below zero in 9 cells, in 0 of 4 GPT-2, 0 of 3 Qwen3 base and 1 of 3 post-trained checkpoints. Index numbers refer to Table A7, in the order of that table, with the kind of change named at the left; the purple frame marks the row and the column of the paper's wording and the grey diagonal, where the fitted and the scored wording are the same, is not coloured. Cells at or below the noise expected from test pairs alone for that checkpoint (the noise column of Figure A5) are stippled; the 9 cells at or below zero (all on Qwen3-4B, lowest −0.12) carry a dot; the white ticks on the colour bar mark the noise range of each role, 0.093 to 0.102 on GPT-2, 0.059 to 0.081 on Qwen3 base and 0.081 to 0.109 on post-trained, and 65 cells fall there. On the bar, the Qwen3 base maximum and post-trained minimum coincide at 0.081.

Table A7: The wordings of the sentiment template. Index numbers are those of Figure A5 and Figure A6. The template is the user text, with the stimulus in place of {stimulus} and the concept word in place of {concept}, and the symbol \n marks a line break. The wording that differs from the paper's only in a space token after the assistant tag is left out of the figures and of this table.

No.	Wording	Kind of change	Template
1	paper	none, the wording of the paper	Consider the amount of {concept} in the following:\n{stimulus}\nThe amount of {concept} is
2	no template	template removed, chat tags kept	{stimulus}
3	bare	template and chat tags removed	{stimulus}
4	no chat tags	chat tags removed, template kept	Consider the amount of {concept} in the following:\n{stimulus}\nThe amount of {concept} is
5	neutral frame	frame that does not name the concept	Consider the following text:\n{stimulus}
6	no trailing question	closing question removed	Consider the amount of {concept} in the following:\n{stimulus}
7	paraphrase 1	paraphrase of the instruction	Rate how much {concept} the following text expresses:\n{stimulus}\nThe amount of {concept} is
8	paraphrase 2	paraphrase of the instruction	How much {concept} is there in this text? \n{stimulus}\nThe amount of {concept} is
9	paraphrase 3	paraphrase of the instruction	Read the text below and assess its {concept}.\n{stimulus}\nIts {concept} is
10	scaffold 1	closing sentence unrelated to the concept	{stimulus}\nConclusion: This movie is
11	scaffold 2	closing sentence unrelated to the concept	{stimulus}\nConclusion: This text is
12	scaffold 3	closing sentence unrelated to the concept	{stimulus}\nConclusion: Overall, it was
13	concept swap 1	concept word replaced by an unrelated one	Consider the amount of furniture in the following:\n{stimulus}\nThe amount of furniture is
14	concept swap 2	concept word replaced by an unrelated one	Consider the amount of geometry in the following:\n{stimulus}\nThe amount of geometry is
15	concept swap 3	concept word replaced by an unrelated one	Consider the amount of weather in the following:\n{stimulus}\nThe amount of weather is
16	concept swap 4	concept word replaced by an unrelated one	Consider the amount of cooking in the following:\n{stimulus}\nThe amount of cooking is

Appendix E Truth and honesty

These support the finding that a truth reader does not beat words alone on TruthfulQA: Figure A7 gives the reader on statements and their negations, Table A8 the readers and the honesty transfer, and Table A9 the reader on all nine Geometry of Truth sets.

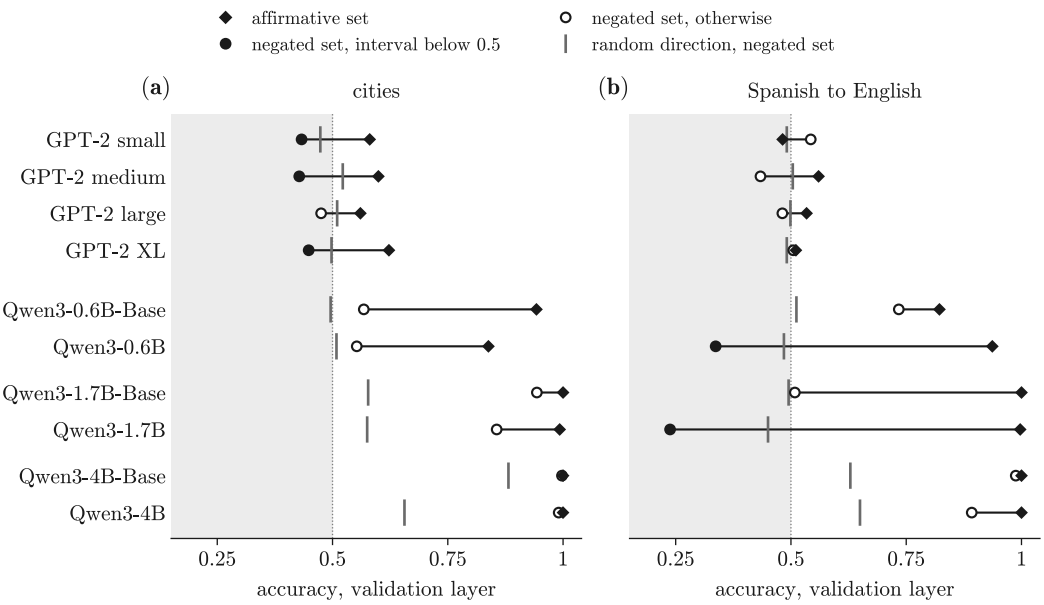


Figure A7: Statements with negation pull a truth reader below chance on some checkpoints: the accuracy on a negated set has a 95 percent Wilson interval wholly below 0.5 for at least one set on 3 of 4 GPT-2, 0 of 3 Qwen3 base and 2 of 3 post-trained checkpoints. Accuracy of the truth reader, at the layer chosen on validation, on statements about cities (a) and on Spanish to English translations (b): a dumbbell from the affirmative set (diamond) to its negation (circle, filled where the Wilson interval of the negated set lies wholly below 0.5), with the accuracy of a random direction on the negated set as a grey tick and the region below chance shaded. Whiskers are not drawn; the criterion uses 95 percent Wilson intervals over the evaluation pairs of each set. GPT-2 reads affirmative cities at only 0.56 to 0.62, against 0.48 to 0.52 for its random direction, so small excursions there are not inversions of a strong reader. On negated cities the reader falls below the random direction by point estimate on 4 of 4 GPT-2 sizes. The truth reader on TruthfulQA is in Figure 8.

Table A8: Truthfulness and honesty readings on every checkpoint, with the baselines of the same questions. The four leftmost columns are TruthfulQA MC1 on 173 test questions for readers fitted on TruthfulQA, on ARC and on the paper's honesty persona prompts, each at the layer chosen on validation questions, and for the model's own answer log-probability; a 95 percent Wilson interval at this size is at most ±0.074 wide. A classifier on the question and answer words reaches 0.676 [0.603, 0.742], one on the characters 0.671 [0.597, 0.736], and chance is 0.228; the last three rows give these baselines, which apply to each of the four TruthfulQA columns. Honesty in template is the accuracy on held-out persona prompts, 1.000 at layer 1 on 4 of 4 GPT-2, 2 of 3 Qwen3 base and 3 of 3 post-trained checkpoints and at layer 2 on Qwen3-4B-Base because the members of a pair differ in the persona instruction; it is not a held-out measure of honesty. In the template the random direction at the same layer reaches 0.714 to 0.832 (GPT-2), 0.606 to 0.617 (Qwen3 base) and 0.610 to 0.657 (post-trained) across checkpoints, per checkpoint in Table A4. Honesty transfer scores the honesty reader on plain true and false statements (59 test pairs, layer chosen on 53 validation pairs) next to the random direction at the same layer. Seed agreement is the mean absolute cosine between the honesty directions of different seeds at relative depth 0.5, not at the transfer layer.

Model	TruthfulQA MC1				Honesty in template			Honesty transfer		Seed agreement
	TruthfulQA reader	ARC reader	Honesty persona reader	Answer log-probability	Accuracy	Layer	Random	Accuracy	Random	Seed agreement
GPT-2 small	0.649	0.383	0.270	0.168	1.000	1	0.832	0.386	0.483	0.986
GPT-2 medium	0.667	0.489	0.476	0.173	1.000	1	0.808	0.598	0.476	0.983
GPT-2 large	0.676	0.408	0.304	0.179	1.000	1	0.763	0.566	0.473	0.984
GPT-2 XL	0.699	0.432	0.371	0.173	1.000	1	0.714	0.492	0.502	0.980
Qwen3-0.6B-Base	0.679	0.406	0.334	0.231	1.000	1	0.617	0.456	0.554	0.975
Qwen3-0.6B	0.644	0.367	0.346	0.324	1.000	1	0.636	0.529	0.512	0.980
Qwen3-1.7B-Base	0.703	0.400	0.443	0.197	1.000	1	0.609	0.778	0.492	0.979
Qwen3-1.7B	0.672	0.407	0.558	0.295	1.000	1	0.610	0.781	0.485	0.979
Qwen3-4B-Base	0.751	0.434	0.511	0.254	1.000	2	0.606	0.658	0.531	0.981
Qwen3-4B	0.741	0.545	0.575	0.318	1.000	1	0.657	0.758	0.505	0.979
Words classifier	0.676									
Characters classifier	0.671									
Chance	0.228									

Table A9: Reader accuracy on every truth set, next to the random direction. Each cell is the accuracy of the reader at the layer chosen on validation pairs, with the accuracy of the random direction at that layer in parentheses. A star marks a reader whose 95 percent Wilson interval lies below 0.5, that is, a reader that gets the label reversed on that set.

Model	cities	negated cities	Spanish to English	negated Spanish to English	larger than	smaller than	cities conjunction	cities disjunction	common claims
GPT-2 small	0.581 (0.523)	0.433* (0.474)	0.482 (0.493)	0.543 (0.491)	0.477 (0.468)	0.503 (0.523)	0.542 (0.512)	0.542 (0.514)	0.656 (0.546)
GPT-2 medium	0.599 (0.482)	0.428* (0.522)	0.560 (0.505)	0.434 (0.504)	0.300* (0.470)	0.730 (0.539)	0.545 (0.479)	0.540 (0.490)	0.642 (0.509)
GPT-2 large	0.560 (0.501)	0.475 (0.510)	0.534 (0.508)	0.481 (0.499)	0.336* (0.434)	0.680 (0.573)	0.538 (0.500)	0.521 (0.504)	0.638 (0.527)
GPT-2 XL	0.622 (0.490)	0.448* (0.498)	0.511 (0.514)	0.505 (0.491)	0.454* (0.447)	0.486 (0.541)	0.532 (0.499)	0.530 (0.500)	0.639 (0.519)
Qwen3-0.6B-Base	0.942 (0.510)	0.568 (0.496)	0.822 (0.503)	0.734 (0.512)	0.783 (0.487)	0.902 (0.526)	0.779 (0.529)	0.640 (0.515)	0.695 (0.518)
Qwen3-0.6B	0.838 (0.583)	0.553 (0.509)	0.937 (0.531)	0.337* (0.485)	0.787 (0.507)	0.868 (0.622)	0.666 (0.509)	0.605 (0.503)	0.676 (0.541)
Qwen3-1.7B-Base	1.000 (0.732)	0.943 (0.577)	1.000 (0.727)	0.508 (0.495)	0.997 (0.653)	0.978 (0.795)	0.960 (0.614)	0.852 (0.567)	0.812 (0.550)
Qwen3-1.7B	0.993 (0.574)	0.856 (0.575)	0.997 (0.641)	0.238* (0.450)	0.982 (0.765)	0.868 (0.562)	0.943 (0.620)	0.780 (0.530)	0.791 (0.551)
Qwen3-4B-Base	1.000 (0.889)	0.998 (0.882)	1.000 (0.725)	0.988 (0.629)	1.000 (0.845)	1.000 (0.814)	0.992 (0.736)	0.960 (0.711)	0.871 (0.662)
Qwen3-4B	1.000 (0.704)	0.991 (0.656)	1.000 (0.808)	0.892 (0.650)	1.000 (0.708)	1.000 (0.593)	0.990 (0.680)	0.929 (0.551)	0.839 (0.577)

Appendix F Steering details

Table A10 gives the values behind the steering finding, including the perplexity of the completions.

Table A10: Sentiment steering at relative coefficient 0.1. Judged sentiment is the mean positive-class probability of SiEBERT over the completions, with a DistilBERT SST-2 classifier as a check in the data file; the instruction columns give the score under a negative or a positive instruction instead of steering. Neutral-text perplexity is the perplexity of 200 held-out WikiText-103 passages under the steered model as a ratio to the unsteered model; completion perplexity is the perplexity of the completions themselves as a ratio to that of the unsteered completions; distinct-1 is the share of distinct tokens in the completions. The columns at the two signs of the coefficient are means over the 3 steering vectors; the column at 0 and the two instruction columns are each a single run. The fluency rule of Figure 9 uses the neutral-text ratio (at most 1.5) and distinct-1 (at least 0.15).

	Judged sentiment at relative coefficient			Judged sentiment under an instruction		Neutral-text perplexity ratio		Completion perplexity ratio		Distinct-1	
Model	−0.1	0	+0.1	Negative	Positive	−0.1	+0.1	−0.1	+0.1	−0.1	+0.1
GPT-2 small	0.274	0.575	0.925	0.416	0.776	1.35	1.24	1.27	1.38	0.251	0.242
GPT-2 medium	0.198	0.550	0.889	0.385	0.756	1.54	1.64	1.30	1.72	0.195	0.267
GPT-2 large	0.379	0.635	0.808	0.468	0.765	1.19	1.17	1.32	1.46	0.255	0.265
GPT-2 XL	0.264	0.626	0.844	0.547	0.835	1.43	1.35	1.69	1.79	0.221	0.255
Qwen3-0.6B-Base	0.221	0.625	0.889	0.407	0.909	1.37	1.30	1.50	1.72	0.225	0.274
Qwen3-0.6B	0.300	0.589	0.764	0.544	0.832	1.29	1.32	1.63	1.45	0.183	0.203
Qwen3-1.7B-Base	0.259	0.659	0.794	0.399	0.867	1.36	1.23	0.98	1.44	0.182	0.266
Qwen3-1.7B	0.366	0.624	0.741	0.476	0.885	1.21	1.12	1.43	1.28	0.181	0.215
Qwen3-4B-Base	0.301	0.643	0.779	0.393	0.857	1.17	1.18	1.06	1.23	0.214	0.290
Qwen3-4B	0.579	0.635	0.666	0.399	0.835	1.25	1.03	0.98	1.13	0.210	0.219

Appendix G Knowledge

These support the finding on known and unknown questions: the transfer between question types on Qwen3-4B-Base (Figure A8), the values of the main figures (Table A11) and every dose curve (Figure A9, Figure A10) behind Figure 6.

The direction moves unknown questions more than known questions on 3 of 3 GPT-2 runs, 3 of 3 Qwen3 base runs and 4 of 6 post-trained runs, by the gap between the absolute mean slopes over the coefficient grid, whatever the sign. A direction counts as above a random direction when the 95 percent interval of that gap lies above zero; with 3 random directions a direction passes by chance with probability at most 1/4. By the largest shift of the mean dose curve, which a slope does not show, known questions move at least as much as unknown ones on 2 of 3 GPT-2 runs, 2 of 3 Qwen3 base runs and 0 of 6 post-trained runs.

The confidence of the model stacks five output signals, and only the pooled level of the stack is stored, so the single probability of the model's own answer token stands for it in Figure 5; no random direction is drawn there because its within-relation level is not stored. Its intervals use the Hanley and McNeil approximation. By point estimate the direction is above the strongest of popularity and the two text classifiers on 3 of 3 GPT-2, 3 of 3 Qwen3 base and 3 of 3 post-trained checkpoints. A supervised probe, fitted on the correct and wrong labels of the training questions (the direction is not), adds 0.033 to 0.048 on GPT-2, 0.020 to 0.044 on Qwen3 base and 0.016 to 0.021 on post-trained checkpoints to the AUROC for predicting wrong answers, with an interval above zero on 4

of 4 GPT-2, 3 of 3 Qwen3 base and 3 of 3 post-trained checkpoints, while the direction changes it by −0.009 to 0.011, −0.001 to 0.002 and −0.001 to 0.001 on the same three groups, with the largest upper end of its interval at 0.032, 0.006 and 0.004. GPT-2 small has no known-against-unknown contrast, so it has no row in the figure; its probe adds 0.037.

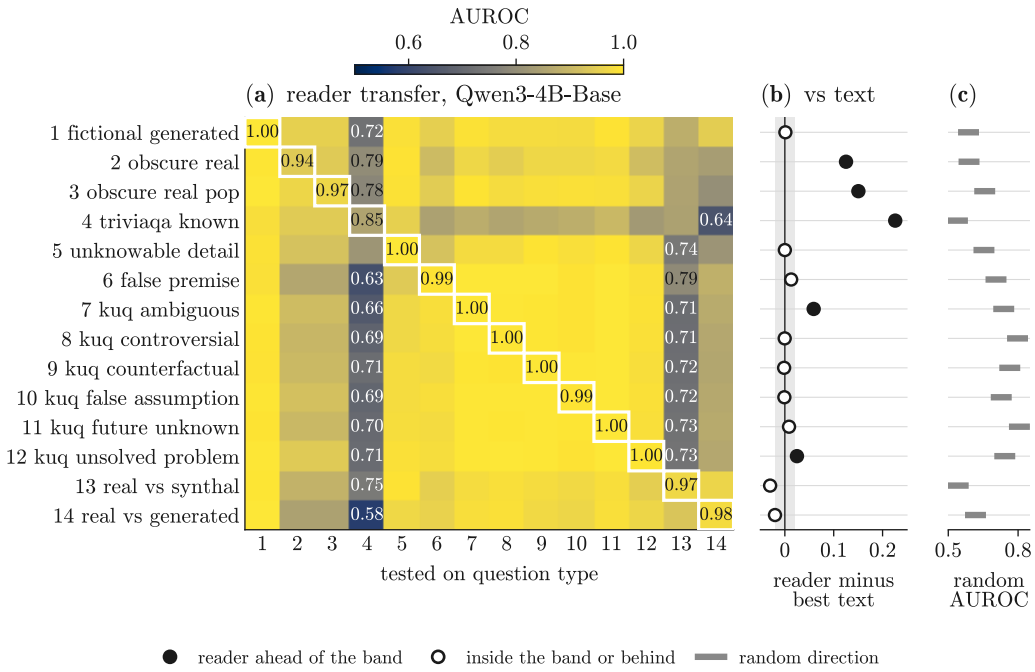


Figure A8: A classifier on the words or characters alone comes within 0.020 of the reader or beats it on 9 of 14 question types. (a) AUROC of a logistic reader fitted on the question type in the row and tested on the type in the column, for Qwen3-4B-Base at its validation layer; the diagonal is outlined in white, and a number is written only on the diagonal and in cells below 0.8. (b) The AUROC of the reader on its own type minus that of the best text classifier (the better of the one on the words and the one on the characters of the questions); the band marks a difference of 0.020 either way, a filled marker is a reader ahead of the band (5 of 14 types) and an open one is a reader inside it or behind. (c) The AUROC of a random direction on the same type, as a short grey bar. A high value in a column of (a) does not show that the direction reads knowledge where a text classifier comes this close.

Table A11: Within-relation AUROC for PopQA questions the model answers correctly against those it does not, for the knowledge direction, the popularity of the subject and classifiers on the words and the characters of the question. Popularity is the monthly Wikipedia page views of the subject [23]; the difference from popularity carries the Hanley and McNeil half-width [33], with an asterisk where it excludes zero. The added columns give the change in AUROC for predicting wrong PopQA answers when the direction or a supervised probe is added to the confidence of the model, which stacks five output signals, with the stored paired bootstrap interval over the test questions. The steering columns count the runs whose direction moves unknown questions more than each of 3 random directions (distinguishable) and more than known questions (selective). GPT-2 small has too few known questions for a contrast.

	Within-relation AUROC				Difference	Added to the model's confidence		Steering runs	
Model	Direction	Popularity	Words	Characters	Direction − popularity	+ direction	+ probe	Distinguishable	Selective
GPT-2 small						no contrast	0.037 [0.025, 0.054]	none	none
GPT-2 medium	0.778	0.728	0.556	0.557	0.050 ± 0.116	−0.004 [−0.023, 0.020]	0.042 [0.020, 0.067]	1 of 1	1 of 1
GPT-2 large	0.815	0.787	0.550	0.536	0.028 ± 0.100	−0.009 [−0.020, 0.002]	0.033 [0.017, 0.050]	1 of 1	1 of 1
GPT-2 XL	0.902	0.838	0.511	0.543	0.065 ± 0.072	0.011 [−0.006, 0.032]	0.048 [0.025, 0.076]	1 of 1	1 of 1
Qwen3-0.6B-Base	0.919	0.799	0.678	0.673	0.120 ± 0.069*	0.000 [0.000, 0.000]	0.044 [0.026, 0.063]	1 of 1	1 of 1
Qwen3-0.6B	0.909	0.752	0.619	0.657	0.157 ± 0.081*	0.000 [−0.002, 0.001]	0.021 [0.012, 0.031]	0 of 2	1 of 2
Qwen3-1.7B-Base	0.959	0.879	0.625	0.636	0.080 ± 0.048*	0.002 [−0.002, 0.006]	0.044 [0.028, 0.061]	1 of 1	1 of 1
Qwen3-1.7B	0.893	0.850	0.561	0.602	0.043 ± 0.058	−0.001 [−0.003, 0.002]	0.016 [0.006, 0.026]	1 of 2	1 of 2
Qwen3-4B-Base	0.957	0.905	0.568	0.555	0.051 ± 0.037*	−0.001 [−0.002, 0.001]	0.020 [0.010, 0.031]	1 of 1	1 of 1
Qwen3-4B	0.927	0.863	0.534	0.553	0.063 ± 0.045*	0.001 [−0.001, 0.004]	0.016 [0.009, 0.024]	2 of 2	2 of 2

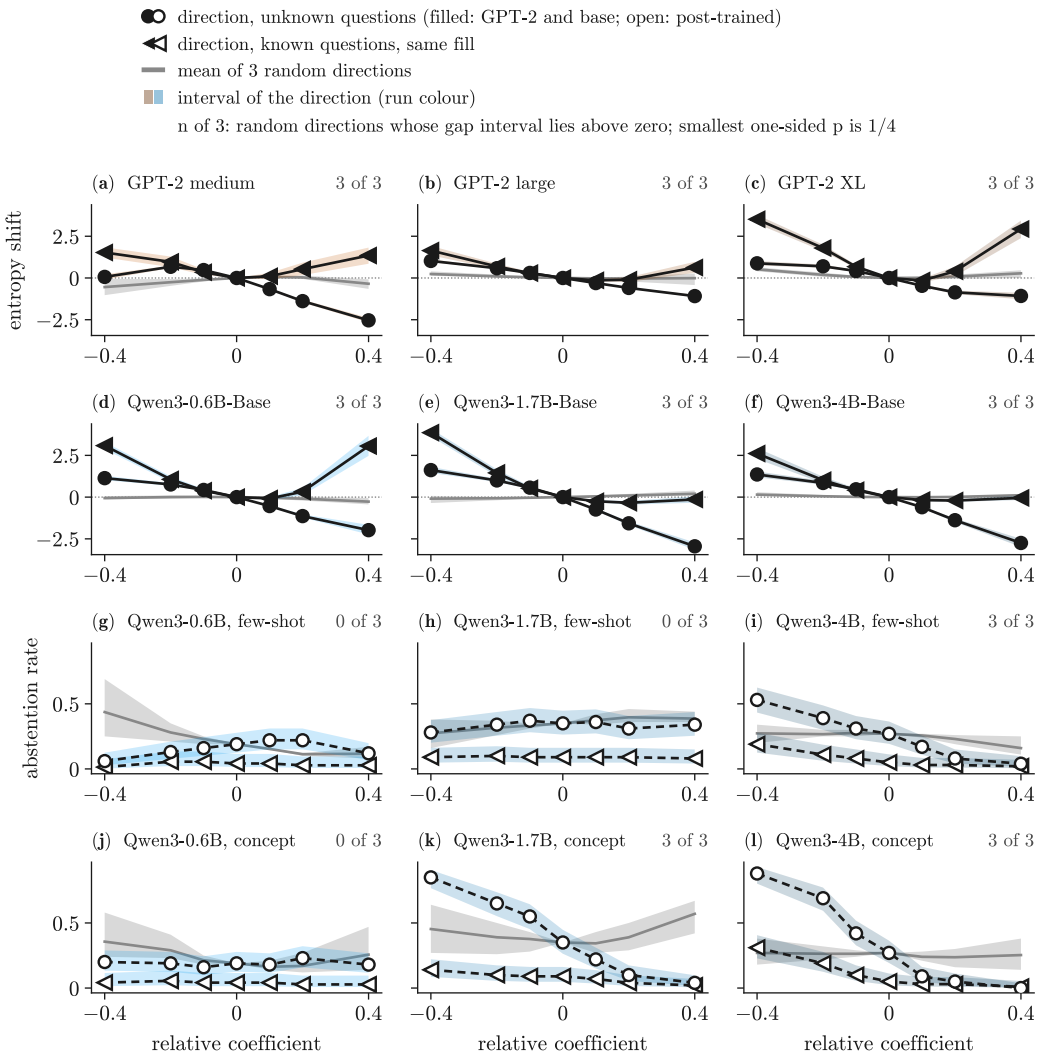


Figure A9: Dose curves of every knowledge-steering run on unknown and known questions. Mean outcome on unknown questions (circle) and known questions (left triangle) against the relative coefficient of the knowledge direction, with 95 percent intervals that resample the questions, and the mean over the 3 random directions on unknown questions with its 95 percent interval that resamples the questions (grey band). Entropy shift is shown for GPT-2 and base checkpoints and abstention rate for post-trained ones, as in Figure 6; rows are GPT-2, Qwen3 base and the post-trained few-shot and concept runs, and panels in the same unit share their vertical range. The tag in each panel is the outcome of Figure 6: the number of the 3 random directions the direction is above. Open markers and dashed lines are post-trained checkpoints.

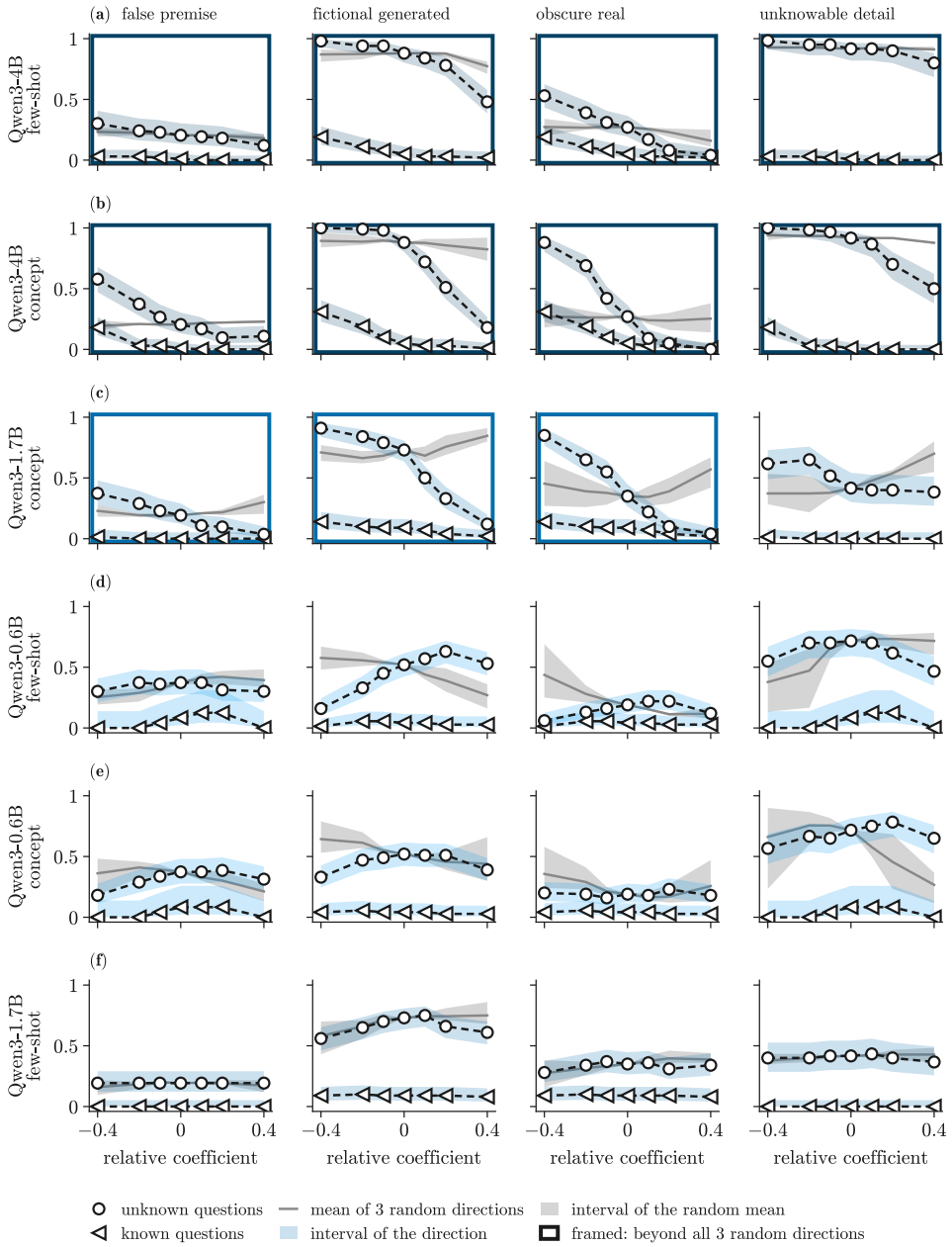


Figure A10: The direction moves the abstention rate of unknown questions more than all 3 random directions in 11 of 24 panels. Abstention rate on unknown questions (circle) and known questions (left triangle) against the relative coefficient of the knowledge direction, for each post-trained run (rows, ordered by the number of framed panels) and question type (columns), with 95 percent intervals that resample the questions, and the mean over the 3 random directions on unknown questions with its 95 percent interval that resamples the questions (grey band). A panel is framed in the colour of its checkpoint when the interval of the difference in the absolute slope on unknown questions between the direction and each of the 3 random directions lies above zero. The test uses each of the 3 random directions, while the plot shows only their mean, so a panel can look closer to the grey band than its test is; with 3 random directions a direction passes by chance with probability at most 1/4. Open markers and dashed lines are post-trained checkpoints; the main view of these runs is Figure 6.

Appendix H Approximate answers

These support the finding on approximate answers. Wikidata gives quantities whose sourced answer is a range (the height of a breed) or a single value (the height of a building); Table A12 gives the values behind Figure 7, which shows what the models answer and what a range reader reads, with the heights of people as a diagnostic. Qwen3 base writes a range on 83 to 99 percent of range items and post-trained on 28 to 68 percent; the highest share on exact items is 15 percent (base) and 0 percent (post-trained), but the range reader is above a classifier on the question words, with the interval of the difference above zero, on 0 of 4 GPT-2, 0 of 3 Qwen3 base and 0 of 3 post-trained checkpoints, sits at the lowest layer on 4 of 4 GPT-2, 2 of 3 Qwen3 base and 3 of 3 post-trained checkpoints, and its random control falls below 0.5, a sign flip between the training and the test concepts, on 0 of 4 GPT-2, 1 of 3 Qwen3 base and 2 of 3 post-trained checkpoints and above 0.95, so the random direction already separates the test concepts, on 3 of 4 GPT-2, 1 of 3 Qwen3 base and 1 of 3 post-trained checkpoints. The control is the mean of random directions whose individual draws are not stored. The reader places the people at 0.871 to 0.991 on GPT-2, 0.890 to 0.943 on Qwen3 base and 0.844 to 0.984 on post-trained checkpoints, next to the breeds, so it does not separate a range-valued answer from a living subject; the position lies outside the random band on 2 of 4 GPT-2, 2 of 3 Qwen3 base and 0 of 3 post-trained checkpoints, but the random bands span -2.316 to 1.472, -2.401 to 0.972 and -2.380 to 15.715 for the same three groups, so the position cannot separate the reader from random. A random band is the range of the random control's position over its 7 training sets, and the random control of the figure and of Table A12 is chosen at its own validation layer, which differs from the layer of the reader on 4 of 4 GPT-2, 3 of 3 Qwen3 base and 2 of 3 post-trained checkpoints. Figure A11 compares the range direction with the knowledge directions.

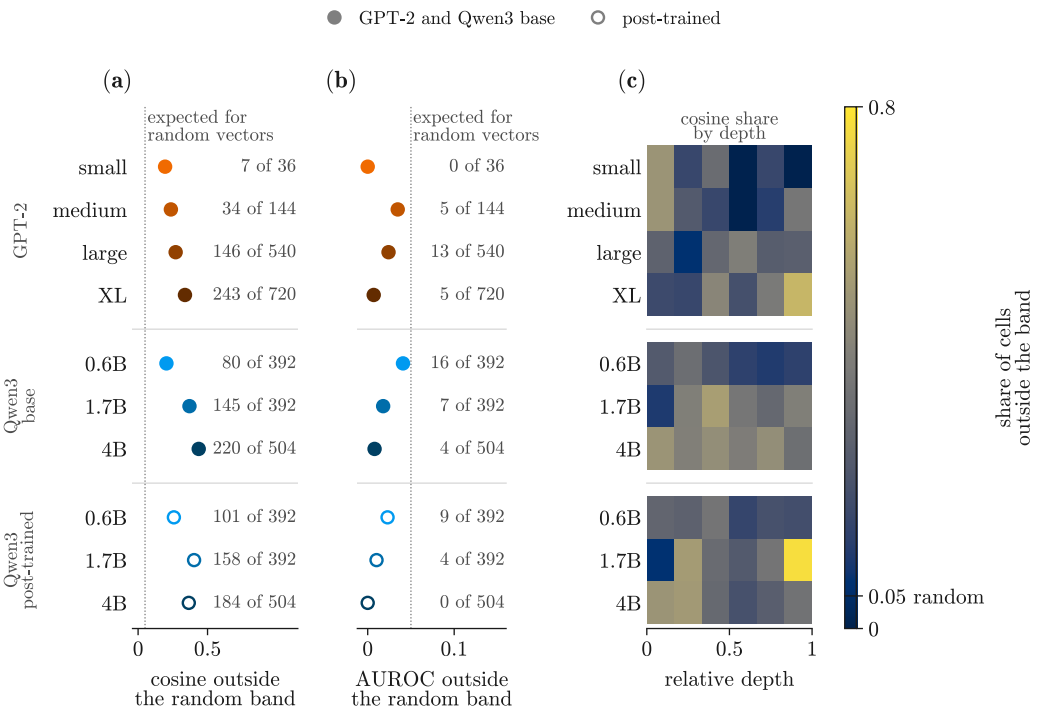


Figure A11: The range direction against the knowledge directions at every layer. Share of the layer-by-type cells whose cosine with the range direction lies outside the band of random directions (a) and whose pooled AUROC interval lies outside the band of random directions (b), one circle per checkpoint in role blocks (filled: GPT-2 and Qwen3 base; open: post-trained), with the number of cells printed at the right of each row; the dotted line is the share expected for random vectors, 0.05. (c) The cosine share by relative depth, in six bins of layers, one row per checkpoint, with 0.05 marked on the colour bar. At layer 1, the layer the plain-template range reader chose on every checkpoint because ties go to the first layer, every cosine is at most 0.132 (GPT-2), 0.116 (Qwen3 base) and 0.115 (post-trained) in absolute value; these values describe that layer and do not locate a link. Over all layers the cosine lies outside its random band in 430 of 1,440 GPT-2 cells, 445 of 1,288 Qwen3 base cells and 443 of 1,288 post-trained cells, and the AUROC interval outside its band in 23, 27 and 13 of the same cells. Each share is a count over few question types (3 to 15 per layer), so it moves in steps, and adjacent layers share a residual stream, so the cells are not independent and 0.05 is only a rough reference.

Table A12: Qwen3 writes a range on 83 to 99 percent of range items (base) and on 28 to 68 percent (post-trained); the highest share on exact items is 15 percent (base) and 0 percent (post-trained), but the range reader is above a classifier on the question words, with the interval of the difference above zero, on 0 of 4 GPT-2, 0 of 3 Qwen3 base and 0 of 3 post-trained checkpoints, is above the 95 percent interval of its random control on 0 of 4 GPT-2, 2 of 3 Qwen3 base and 2 of 3 post-trained checkpoints; the control falls below 0.5 on 0 of 4 GPT-2, 1 of 3 Qwen3 base and 2 of 3 post-trained checkpoints and above 0.95 on 3 of 4 GPT-2, 1 of 3 Qwen3 base and 1 of 3 post-trained checkpoints. The three leftmost columns give the percentage of greedy answers written as a range on range items, exact items and the heights of 120 people. Reader is the AUROC on held-out durations of a range reader fitted on heights and wingspans with the concept prompt, next to its random control (the mean score of 5 random directions oriented on the training labels) and its difference from a classifier on the question words (0.929), with the Hanley and McNeil half-width and an asterisk where it excludes zero. Position places the 120 people between the test buildings (0) and the breeds (1); the random band is the range of the random control's position over its 7 training sets, and the last column says whether the reader's position lies inside it.

Model	Greedy answers written as a range (percent)			AUROC on held-out durations			People between buildings (0) and breeds (1)		
	Range items	Exact items	People	Reader	Random control	Reader − words	Position	Random band	Inside band
GPT-2 small	100	98	100	0.960	0.995	0.031 ± 0.097	0.991	[−2.316, 0.968]	no
GPT-2 medium	98	35	2	0.967	0.967	0.038 ± 0.094	0.894	[0.372, 1.079]	yes
GPT-2 large	44	2	0	0.989	0.996	0.060 ± 0.083	0.871	[−0.275, 0.813]	no
GPT-2 XL	59	0	0	0.989	0.926	0.060 ± 0.083	0.919	[−0.435, 1.472]	yes
Qwen3-0.6B-Base	99	15	0	0.973	1.000	0.043 ± 0.091	0.943	[−2.401, 0.920]	no
Qwen3-0.6B	68	0	0	0.991	0.960	0.062 ± 0.082	0.921	[−0.680, 15.715]	yes
Qwen3-1.7B-Base	84	0	0	1.000	0.949	0.071 ± 0.077	0.890	[−1.942, 0.530]	no
Qwen3-1.7B	28	0	0	1.000	0.194	0.071 ± 0.077	0.984	[0.624, 2.462]	yes
Qwen3-4B-Base	83	0	0	0.667	0.346	−0.263 ± 0.173*	0.903	[−0.161, 0.972]	yes
Qwen3-4B	54	0	0	0.998	0.080	0.069 ± 0.078	0.844	[−2.380, 1.543]	yes

Appendix I Post-training details

Table A13 gives the stored base-against-post-trained comparisons behind the finding that post-training lowers held-out sentiment reading.

Table A13: Stored comparisons of the 0.6B and 1.7B pairs. The difference is post-trained minus base, taken per seed on the same stimuli, with a percentile bootstrap interval over the 10 seeds; each checkpoint is read at its own layer, given as relative depth. Each accuracy is read from pairs with two balanced classes, where chance is 0.500. Each accuracy has the random direction of Table A4 next to it, read from the same run at the same layer. Figure 10 draws the seed-paired intervals for the 0.6B and 1.7B pairs and, for the 4B pair and the measures without seeds, a conservative interval over the test items.

Measure	Size	Base	Base random	Post-trained	Post-trained random	Difference	Interval	Base depth	Post-trained depth
utility	0.6B	0.744	0.628	0.732	0.619	−0.012	[−0.024, 0.003]	0.821	1.000
morality	0.6B	0.793	0.551	0.778	0.551	−0.016	[−0.031, −0.004]	0.857	0.643
truth	0.6B	0.763	0.527	0.736	0.542	−0.027	[−0.066, 0.008]	0.821	0.643
honesty, transfer	0.6B	0.456	0.554	0.529	0.512	0.073	[0.059, 0.086]	0.964	0.750
utility	1.7B	0.768	0.591	0.756	0.612	−0.012	[−0.023, −0.001]	0.643	1.000
morality	1.7B	0.896	0.582	0.892	0.700	−0.004	[−0.010, 0.002]	0.714	0.643
truth	1.7B	0.934	0.680	0.937	0.598	0.003	[−0.015, 0.020]	0.643	0.643
honesty, transfer	1.7B	0.778	0.492	0.781	0.485	0.003	[−0.042, 0.049]	0.714	0.714

Appendix J Other experiments

Further experiments ran on the same checkpoints and are reported in the findings document of the repository. They have no figure or table on this page: the LoRRA adapter and its TruthfulQA control, emotion reading and emotion steering, the lie gallery with honesty and lie steering, template steering, the logit lens and token position of the reader. Each link opens the section of the findings document that reports the experiment; this page makes no claim about what they show. Read them as a record of what was run, not as results of this page.

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